RWTH AACHEN UNIVERSITY

Computer Vision

Computer Vision – Lecture 16

Epipolar Geometry & Stereo Basics

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Announcements

- Lecture evaluation
  - Please fill out the evaluation form

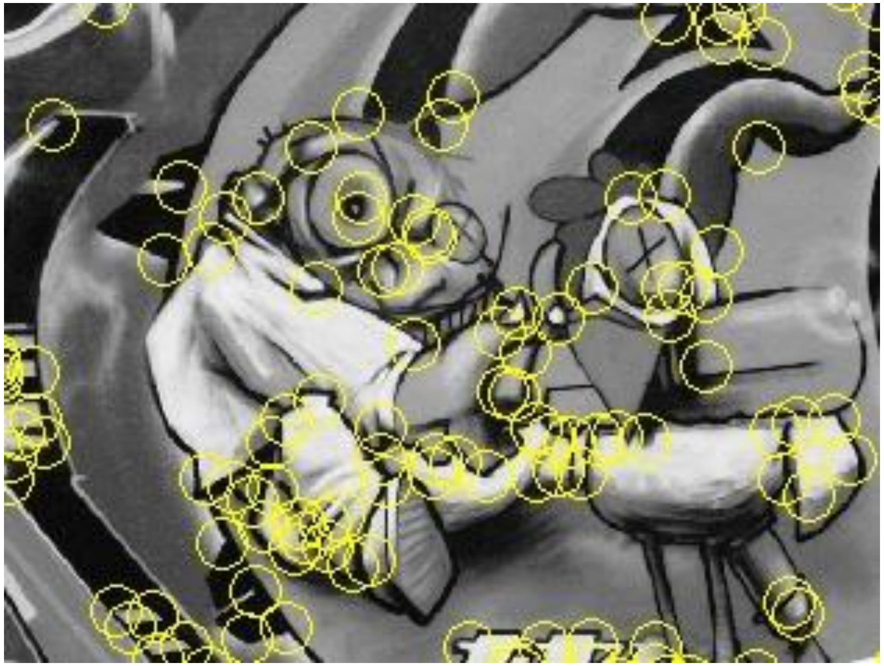
Course Outline

- Image Processing Basics
- Segmentation & Grouping
- Object Recognition & Categorization
- Deep Learning for Vision
- 3D Reconstruction
  - Local Features
  - Epipolar Geometry and Stereo Basics
  - Camera calibration & Triangulation
  - Uncalibrated Reconstruction & Structure-from-Motion

Topics of This Lecture

- Keypoint Localization
  - Recap: Interest points
  - Harris detector
- Local Descriptors
  - SIFT descriptor
- Epipolar geometry
  - Depth with stereo
  - Geometry for a simple stereo system
  - Case example with parallel optical axes
  - General case with calibrated cameras
- Stereopsis & 3D Reconstruction
  - Correspondence search
  - Additional correspondence constraints

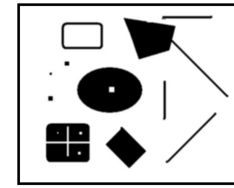
Keypoint Localization



- Goals:
  - Repeatable detection
  - Precise localization
  - Interesting content

$\Rightarrow$ *Look for two-dimensional signal changes*

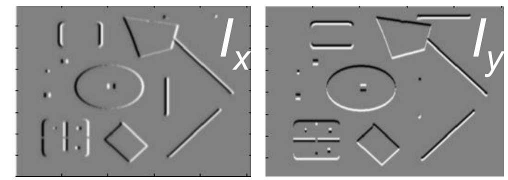
Recap: Harris Detector [Harris88]



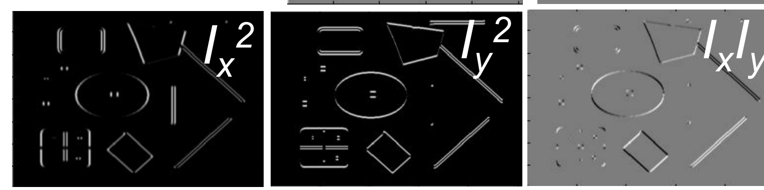
- Compute second moment matrix (autocorrelation matrix)

$$M(\sigma_I,\sigma_D)=g(\sigma_I)*\begin{bmatrix} I_x^2(\sigma_D) & I_xI_y(\sigma_D) \\ I_xI_y(\sigma_D) & I_y^2(\sigma_D) \end{bmatrix}$$

1. Image derivatives



2. Square of derivatives



3. Gaussian filter $g(\sigma_I)$

4. Cornerness function – two strong eigenvalues

$$R=\det[M(\sigma_I,\sigma_D)]-\alpha[\operatorname{trace}(M(\sigma_I,\sigma_D))]^2$$

$$=g(I_x^2)g(I_y^2)-[g(I_xI_y)]^2-\alpha[g(I_x^2)+g(I_y^2)]^2$$

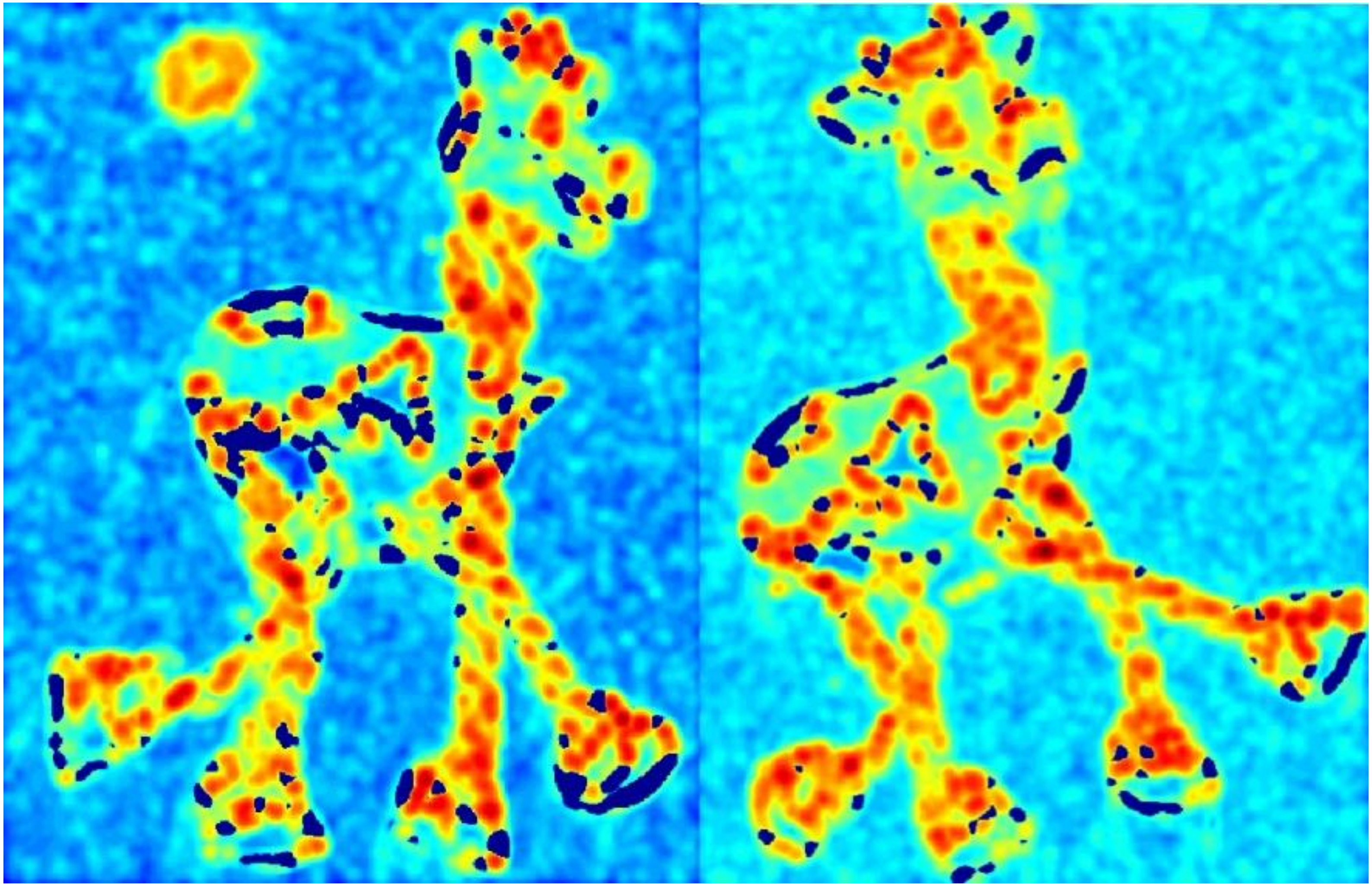
5. Perform non-maximum suppression

Slide credit: Krystian Mikolajczyk

Harris Detector: Workflow

Slide adapted from Darya Frolova, Denis Simakov

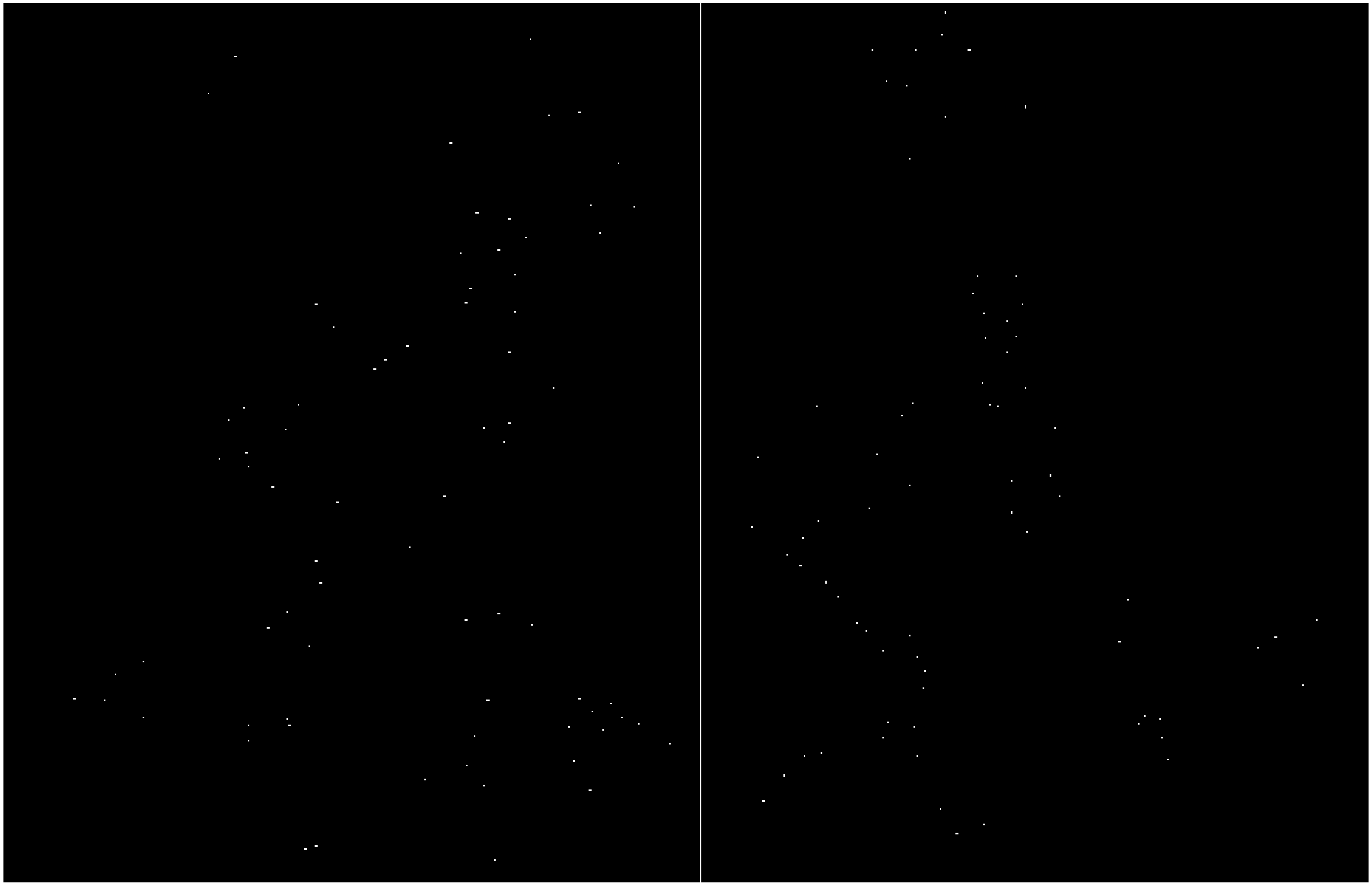
Harris Detector: Workflow



- Compute corner responses R

Slide adapted from Darya Frolova, Denis Simakov

Harris Detector: Workflow



- Take only the local maxima of R, where $R >$ threshold.

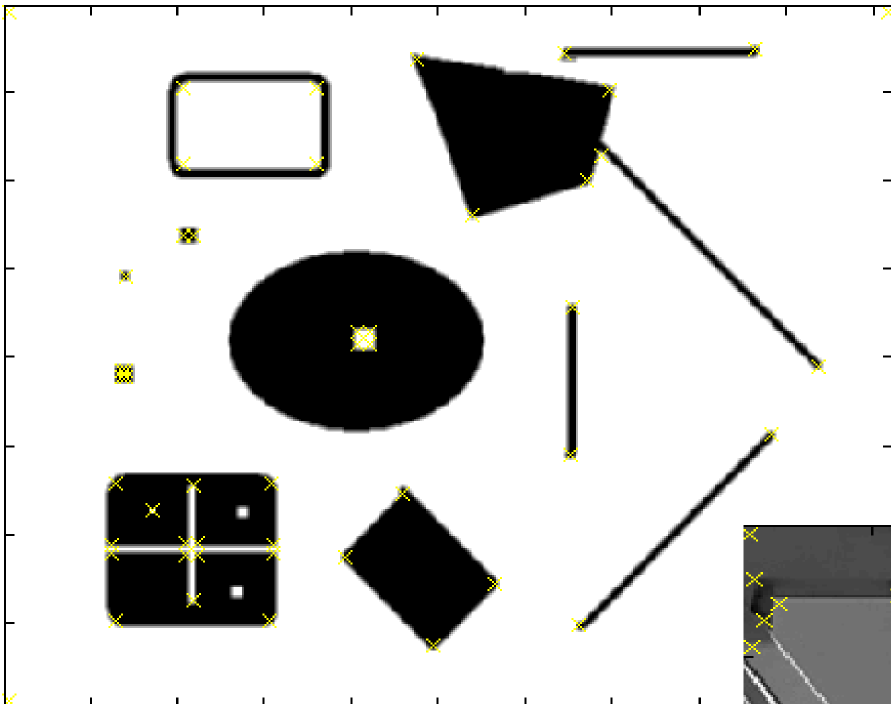
Slide adapted from Darya Frolova, Denis Simakov

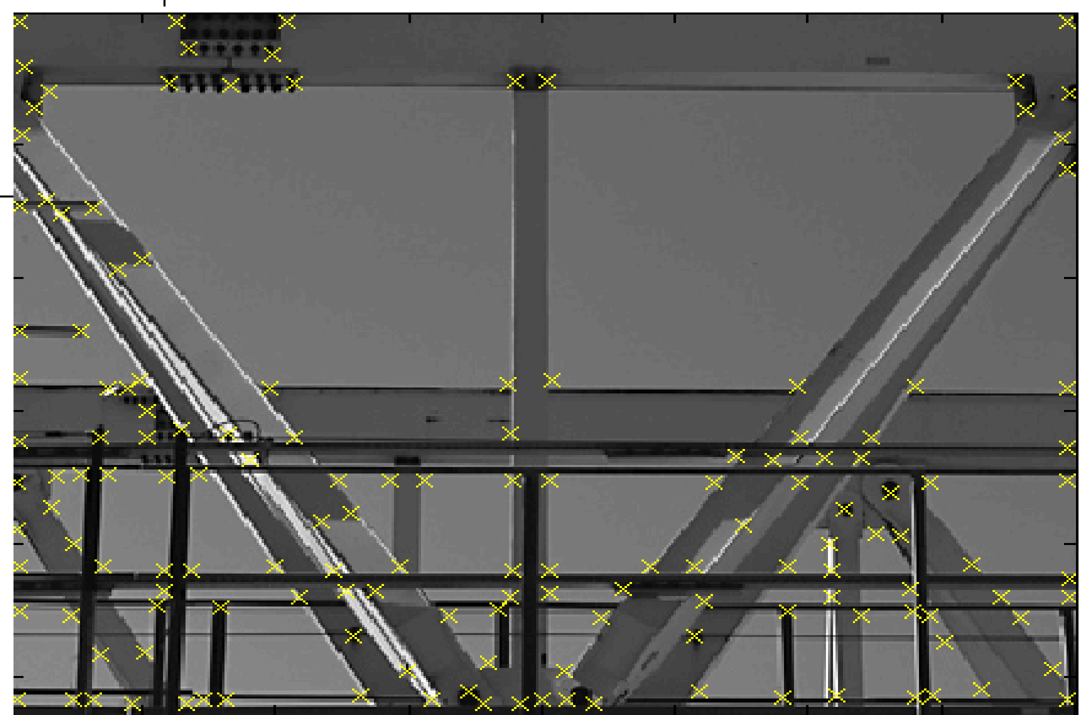
Harris Detector: Workflow

- Resulting Harris points

Slide adapted from Darya Frolova, Denis Simakov

Harris Detector – Responses [Harris88]

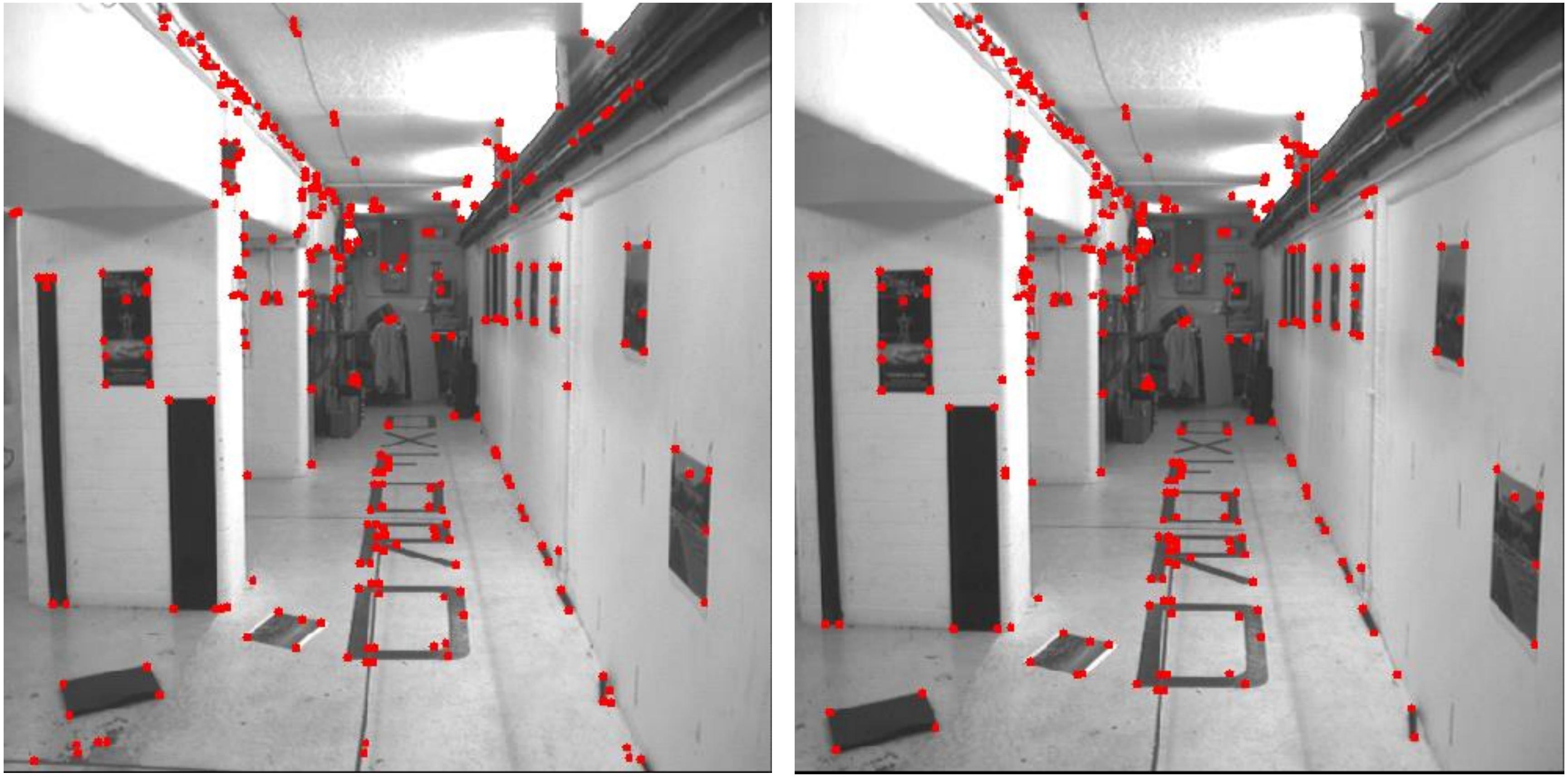




Effect: A very precise corner detector.

Slide credit: Krystian Mikolajczyk

Harris Detector – Responses [Harris88]

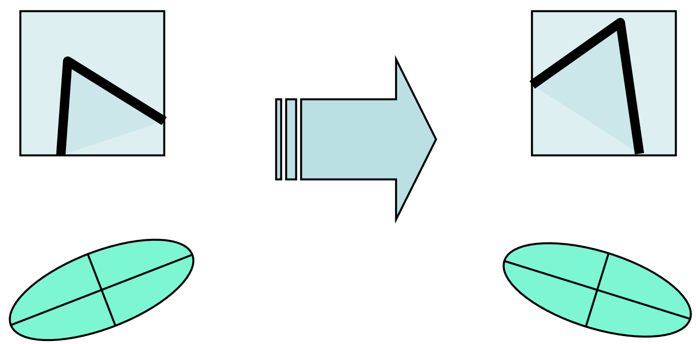


- Results are well suited for finding stereo correspondences

Slide credit: Kristen Grauman

Harris Detector: Properties

- Rotation invariance?



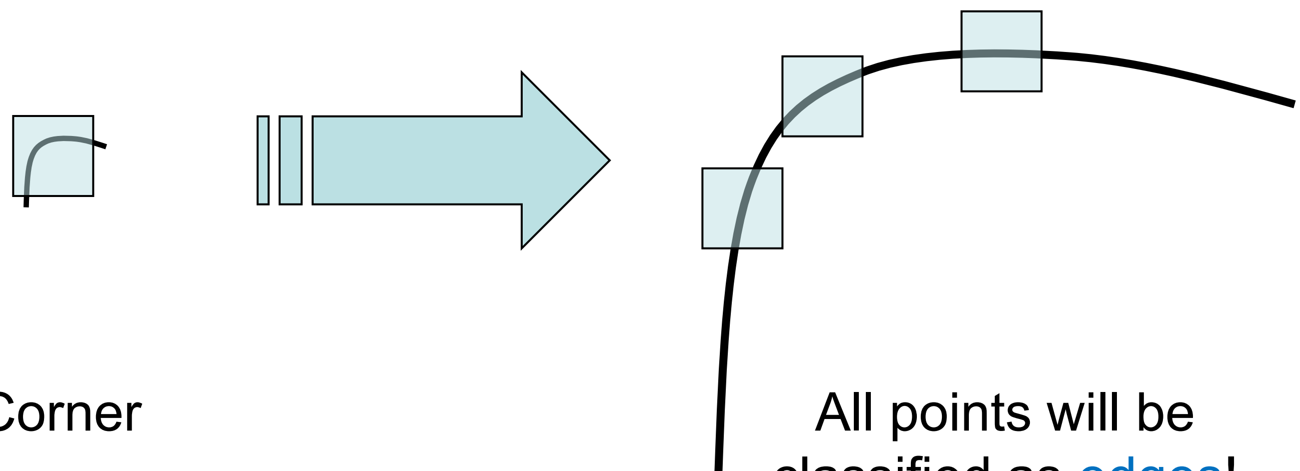
Ellipse rotates but its shape (i.e. eigenvalues) remains the same

Corner response R is invariant to image rotation

Slide credit: Kristen Grauman

Harris Detector: Properties

- Rotation invariance
- Scale invariance?



Corner

All points will be classified as edges!

Not invariant to image scale!

Slide credit: Kristen Grauman

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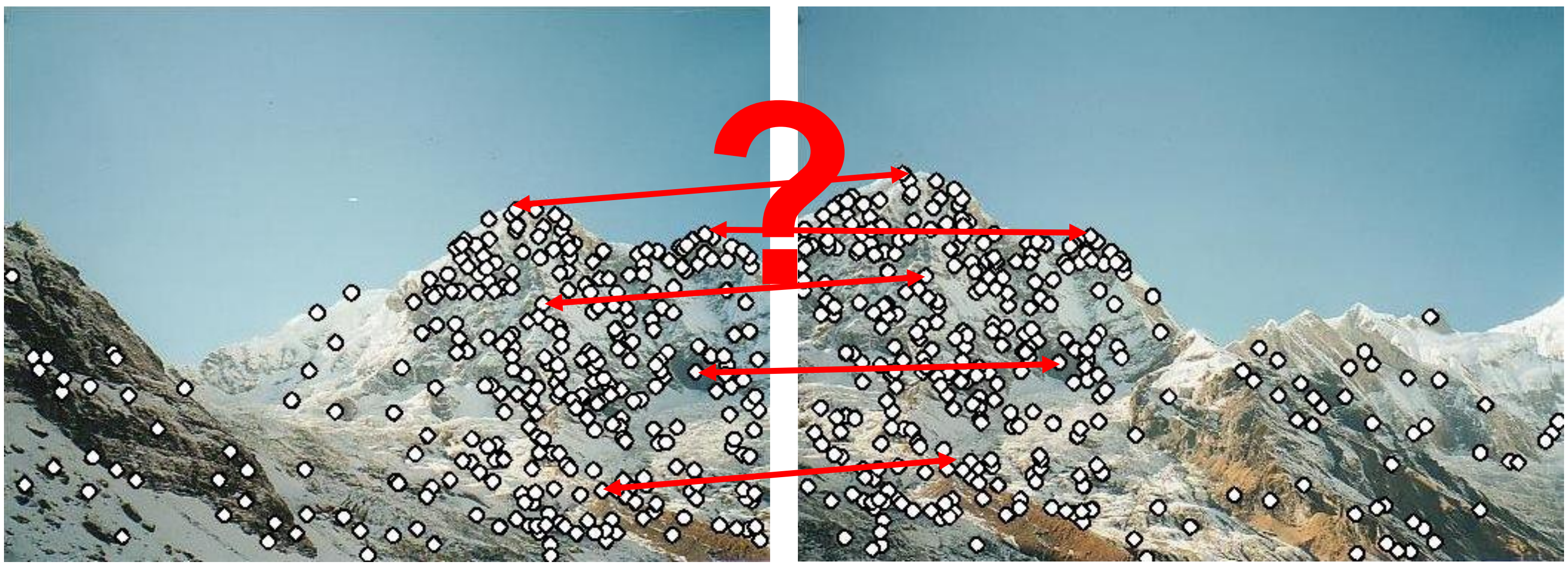
Local Descriptors

- We know how to detect points
- Next question:

How to *describe* them for matching?



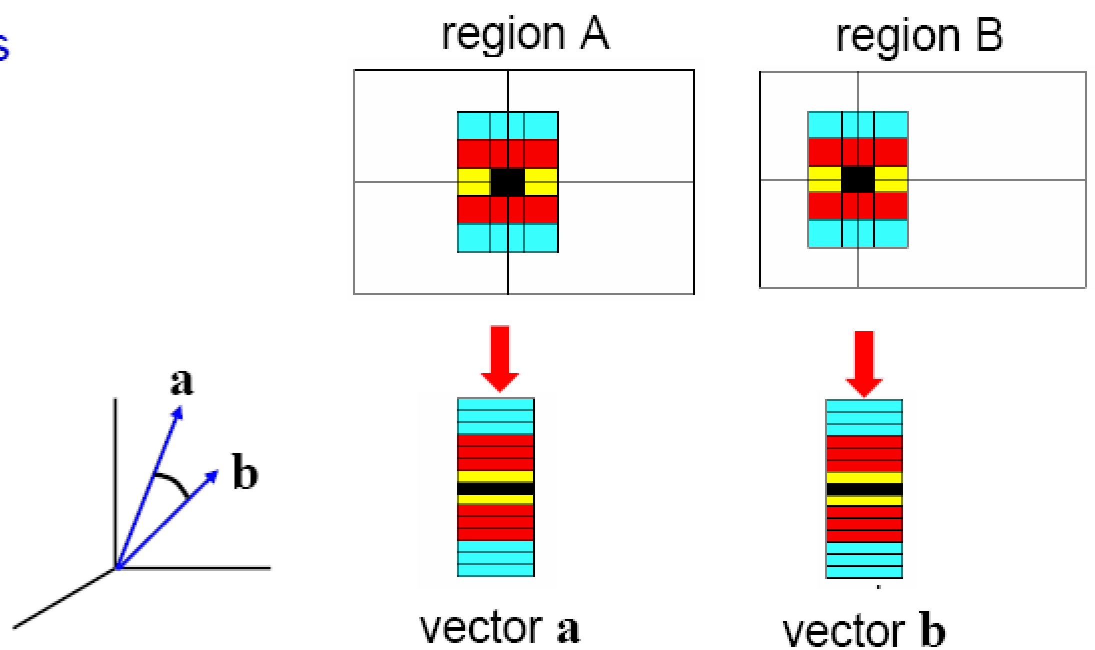
Point descriptor should be:
1. Invariant
2. Distinctive

Local Descriptors

- Simplest descriptor: list of intensities within a patch.
- What is this going to be invariant to?

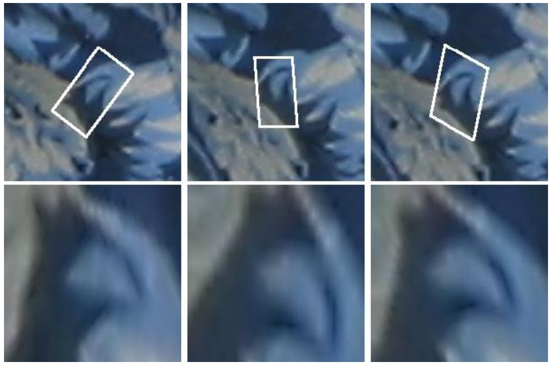
Write regions as vectors

$$A \to \mathbf{a}, \quad B \to \mathbf{b}$$

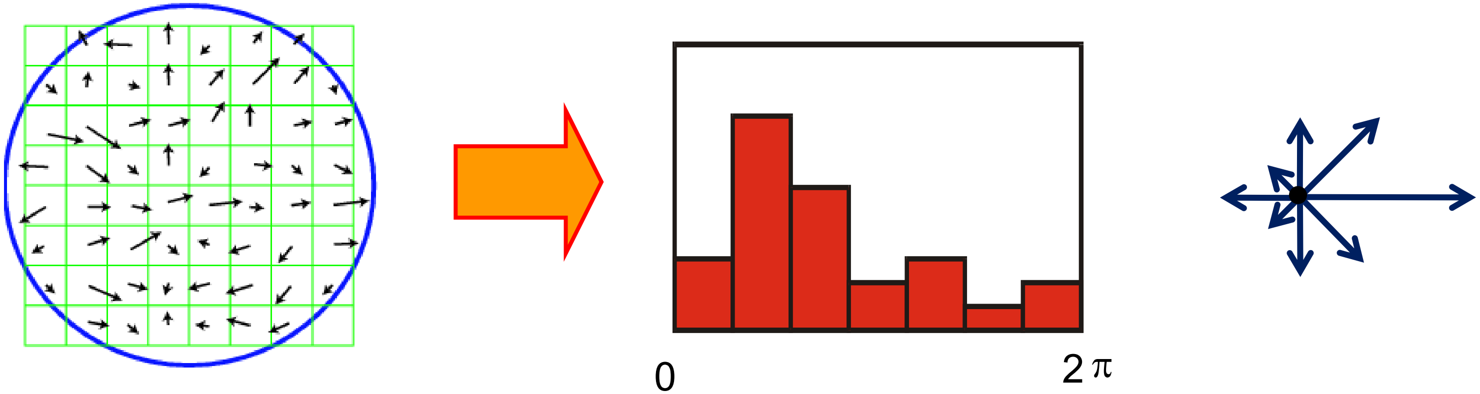


Feature Descriptors

- Disadvantage of patches as descriptors:
  - Small shifts can affect matching score a lot

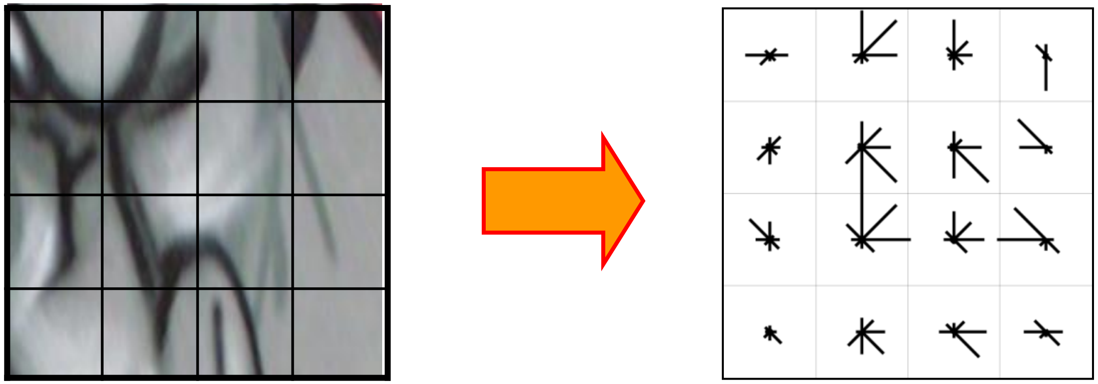


- Solution: histograms



Feature Descriptors: SIFT

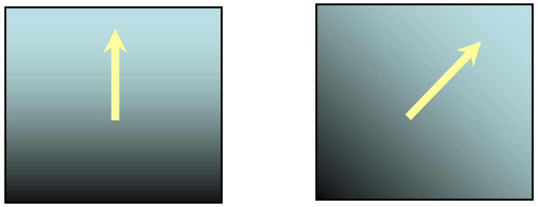
- Scale Invariant Feature Transform
- Descriptor computation:
  - Divide patch into 4x4 sub-patches: 16 cells
  - Compute histogram of gradient orientations (8 reference angles) for all pixels inside each sub-patch
  - Resulting descriptor: 4x4x8 = 128 dimensions



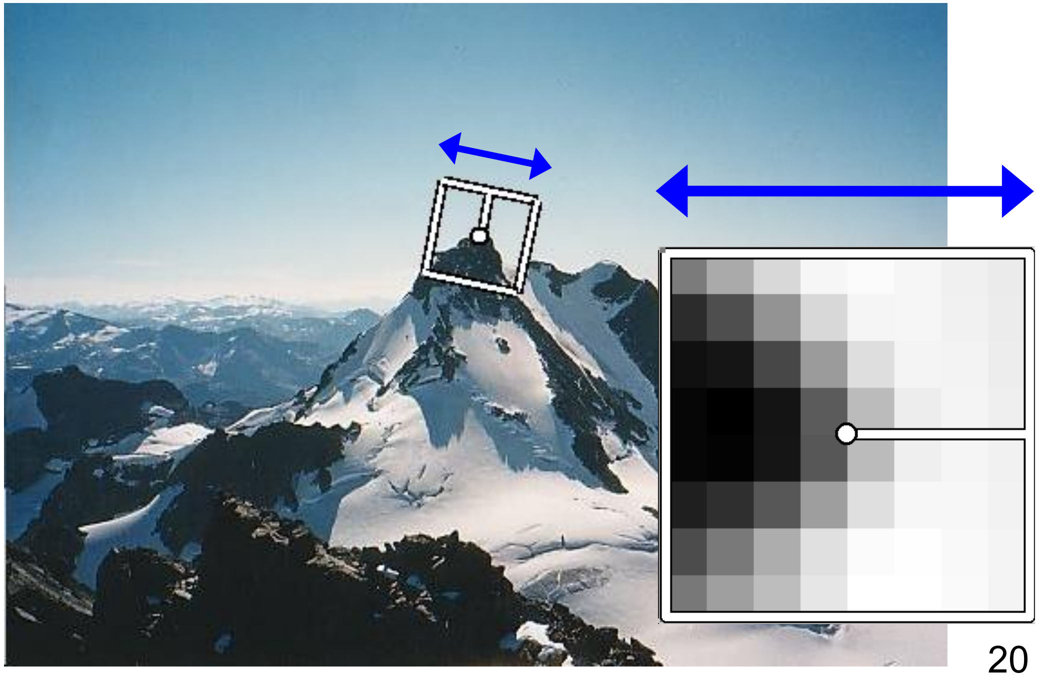
David G. Lowe. "Distinctive image features from scale-invariant keypoints." *IJCV* 60 (2), pp. 91-110, 2004.

Rotation Invariant Descriptors

- Find local orientation
  - Dominant direction of gradient for the image patch



- Rotate patch according to this angle
  - This puts the patches into a canonical orientation.

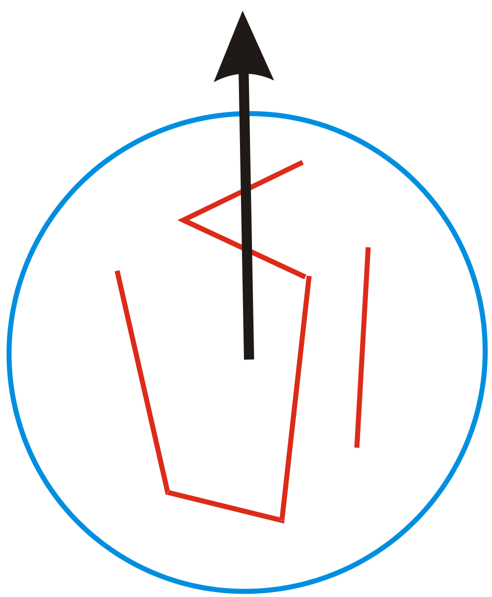


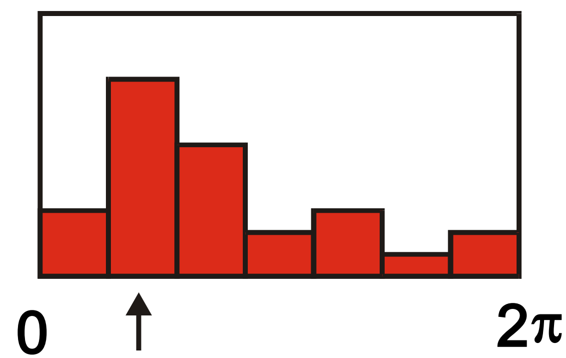
Slide credit: Svetlana Lazebnik, Matthew Brown

Orientation Normalization: Computation

- Compute orientation histogram
- Select dominant orientation
- Normalize: rotate to fixed orientation

[Lowe, 1999]





Slide adapted from David Lowe

Summary: SIFT

- Extraordinarily robust scale invariant feature detector + descriptor
  - Can handle changes in viewpoint up to ~60 deg. out-of-plane rotation
  - Can handle significant changes in illumination
    - Sometimes even day vs. night (below)
  - Fast and efficient—can run in real time
  - Lots of code available
    - http://people.csail.mit.edu/albert/ladypack/wiki/index.php/Known_implementations_of_SIFT


Slide credit: Steve Seitz

Working with SIFT Descriptors

- One image yields:
  - n 128-dimensional descriptors: each one is a histogram of the gradient orientations within a patch
    - [n x 128 matrix]
  - n scale parameters specifying the size of each patch
    - [n x 1 vector]
  - n orientation parameters specifying the angle of the patch
    - [n x 1 vector]
  - n 2D points giving positions of the patches
    - [n x 2 matrix]

You Can Try It At Home…

- For most local feature detectors, executables are available online:
- http://robots.ox.ac.uk/~vgg/research/affine
- http://www.cs.ubc.ca/~lowe/keypoints/
- http://www.vision.ee.ethz.ch/~surf
- http://homes.esat.kuleuven.be/~ncorneli/gpusurf/

Affine Covariant Features

Collaborative work between: the Visual Geometry Group, Katholieke Universiteit Leuven, Inria Rhone-Alpes and the Center for Machine Perception.

Affine Covariant Region Detectors

Input image

→ Region detector

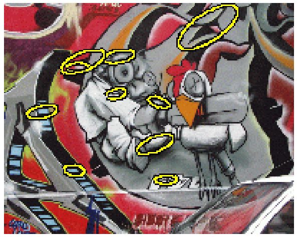
Detector output

format:
1.0
m
$u_1\ v_1\ a_1\ b_1\ c_1$
:
:
$u_m\ v_m\ a_m\ b_m\ c_m$

output example:
img1.haraff

→ display_features.m

Image with displayed regions



Parameters defining an affine region

`u,v,a,b,c` in `a(x-u)(x-u)+2b(x-u)(y-v)+c(y-v)(y-v)=1`
with `(0,0)` at image top left corner

Code

- provided by the authors, see publications for details and links to authors web sites.

Linux binaries	Example of use	Displaying r
Harris-Affine & Hessian-Affine	`prompt>./h_affine.ln -haraff -i img1.ppm -o img1.haraff -thres 1000`	`matlab>> d`
	`prompt>./h_affine.ln -hesaff -i img1.ppm -o img1.hesaff -thres 500`	`matlab>> d`
MSER - Maximaly stable extremal regions (also Windows)	`prompt>./mser.ln -t 2 -es 2 -i img1.ppm -o img1.mser`	`matlab>> d`
IBR - Intensity extrema based detector	`prompt>./ibr.ln img1.ppm img1.ibr -scalefactor 1.0`	`matlab>> d`
EBR - Edge based detector	`prompt> ./ebr.ln img1.ppm img1.ebr`	`matlab>> d`
Salient region detector	`prompt>./salient.ln img1.ppm img1.sal`	`matlab>> d`

http://www.robots.ox.ac.uk/~vgg/research/affine/detectors.html#binaries

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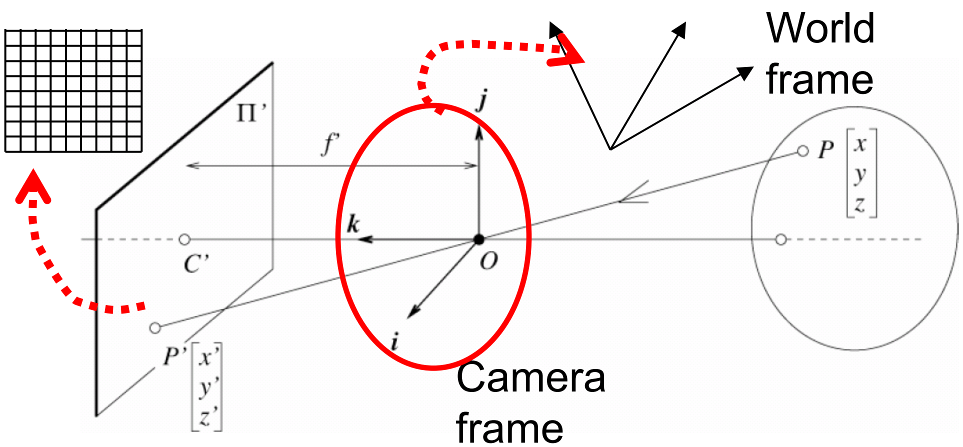
Camera Calibration



Extrinsic parameters:
Camera frame ↔ Reference frame

Intrinsic parameters:
Image coordinates relative to camera ↔ Pixel coordinates

- Parameters
  - *Extrinsic*: rotation matrix and translation vector
  - *Intrinsic*: focal length, pixel sizes (mm), image center point, radial distortion parameters

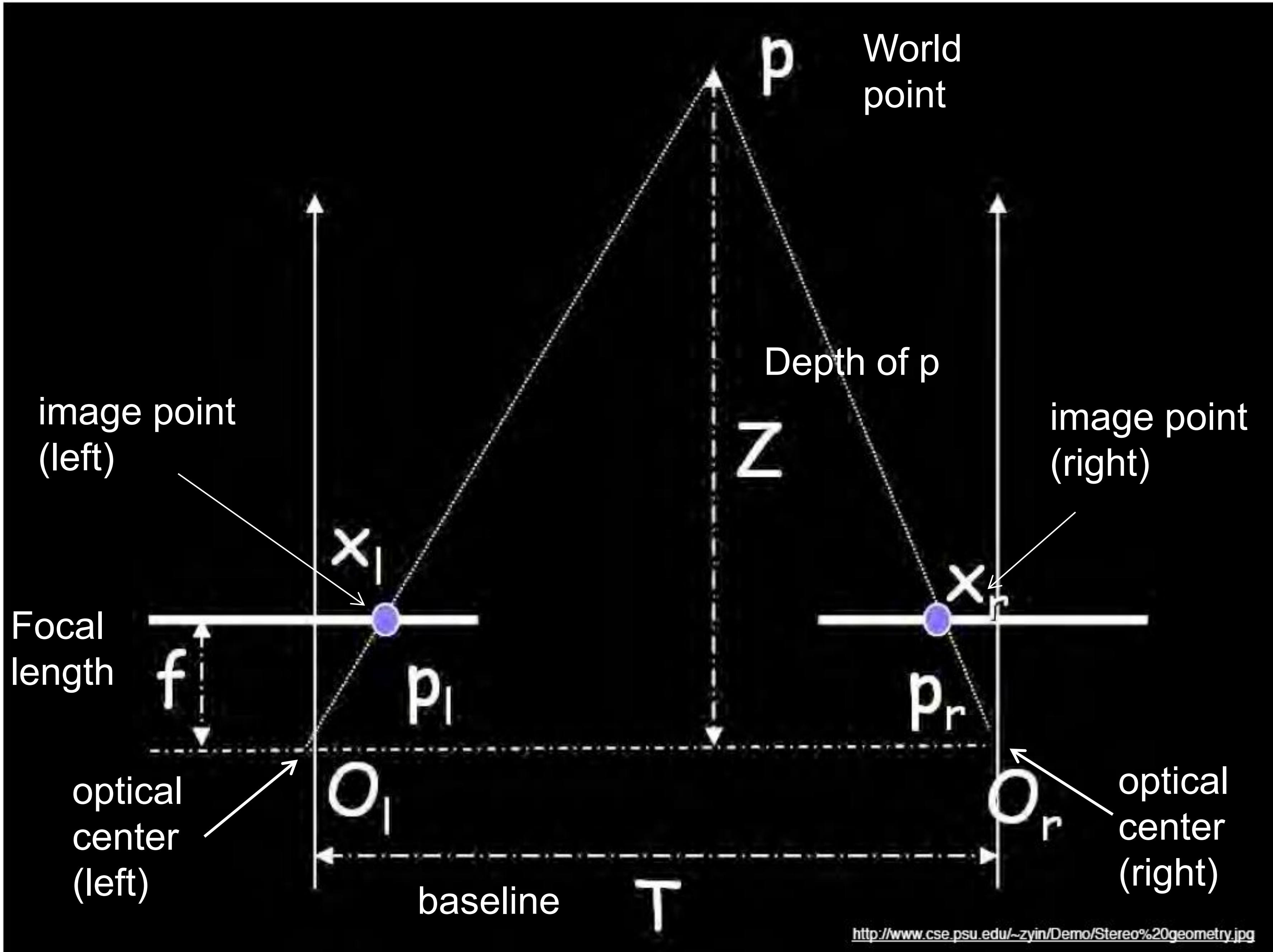
We'll assume for now that these parameters are given and fixed.

Slide credit: Kristen Grauman

Geometry for a Simple Stereo System

- First, assuming parallel optical axes, known camera parameters (i.e., calibrated cameras):

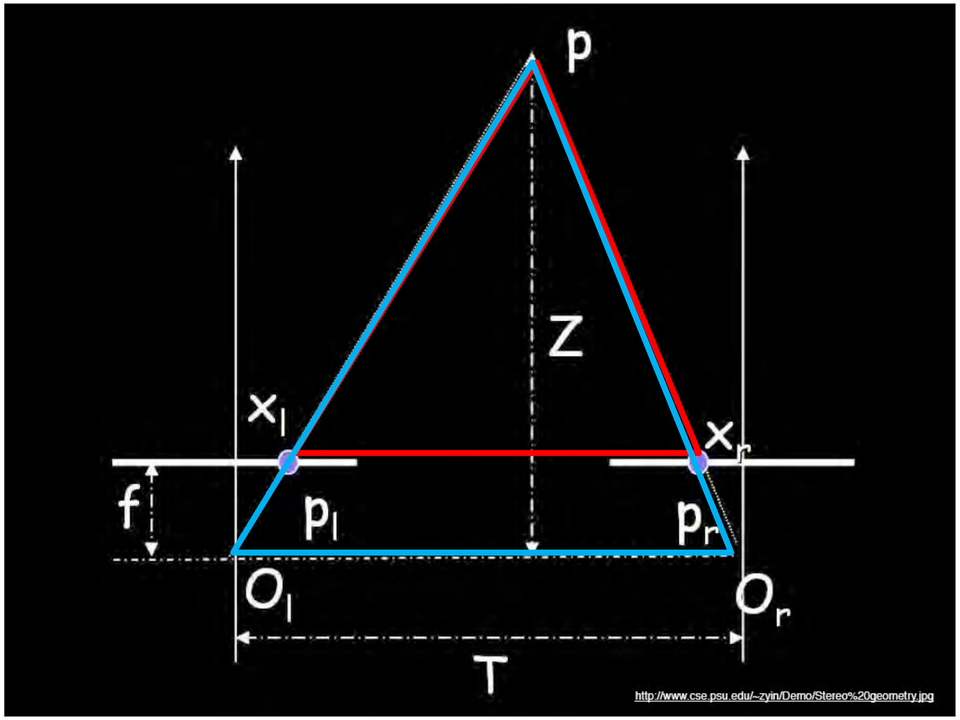
Slide credit: Kristen Grauman



Slide credit: Kristen Grauman

Geometry for a Simple Stereo System

- Assume parallel optical axes, known camera parameters (i.e., calibrated cameras). We can triangulate via:



Similar triangles (p_l, P, p_r) and (O_l, P, O_r):

$$\frac{T - (x_r - x_l)}{Z - f} = \frac{T}{Z}$$

$$Z = f \frac{T}{x_r - x_l}$$

"disparity"

Slide credit: Kristen Grauman

Depth From Disparity

Image $I(x,y)$

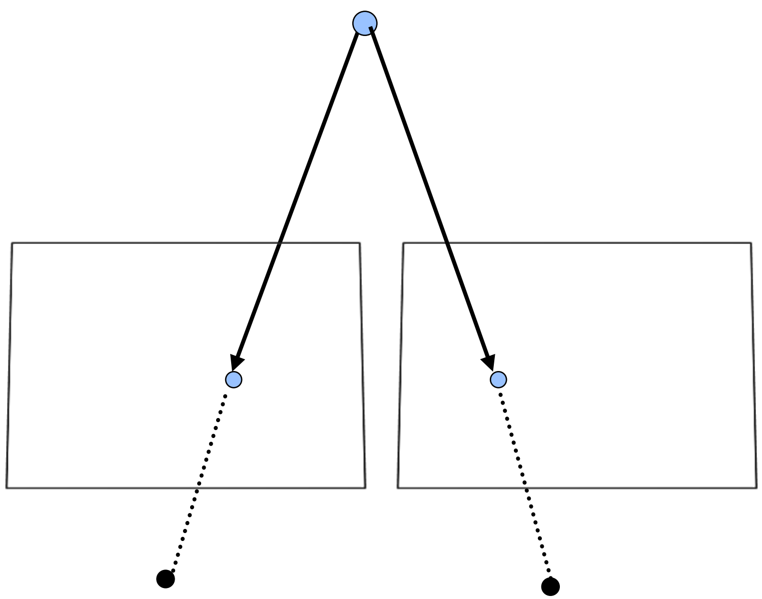
Disparity map $D(x,y)$

Image $I'(x',y')$

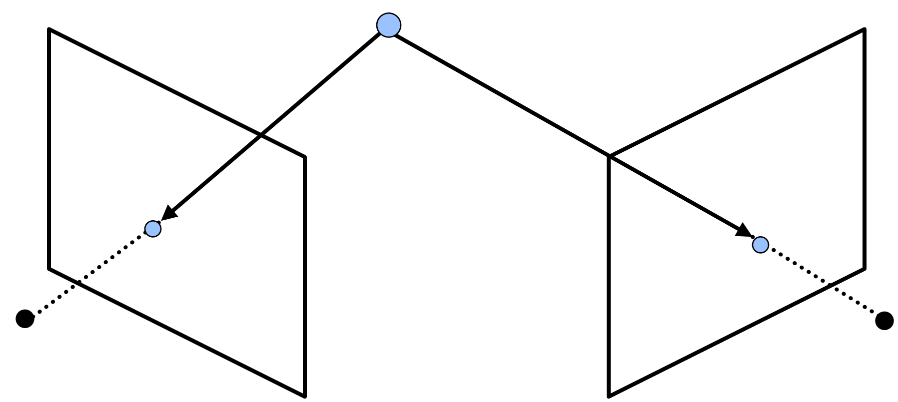
$$(x',y') = (x + D(x,y), y)$$

General Case With Calibrated Cameras

- The two cameras do not need to have parallel optical axes.

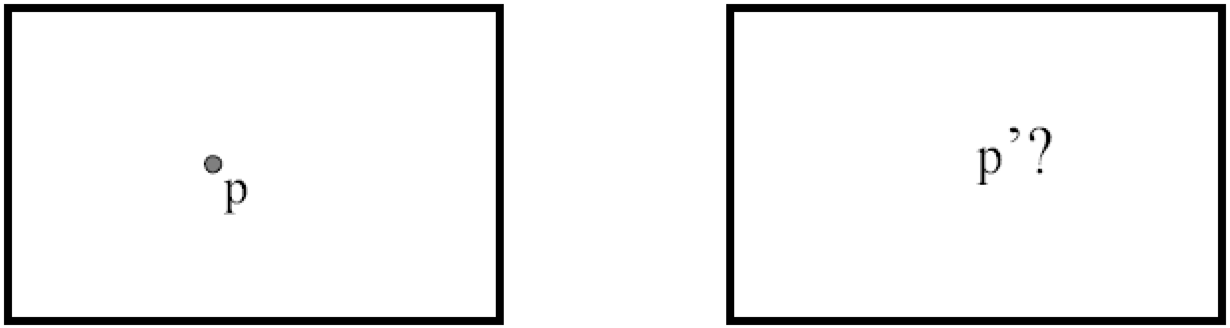


vs.



Slide credit: Kristen Grauman, Steve Seitz

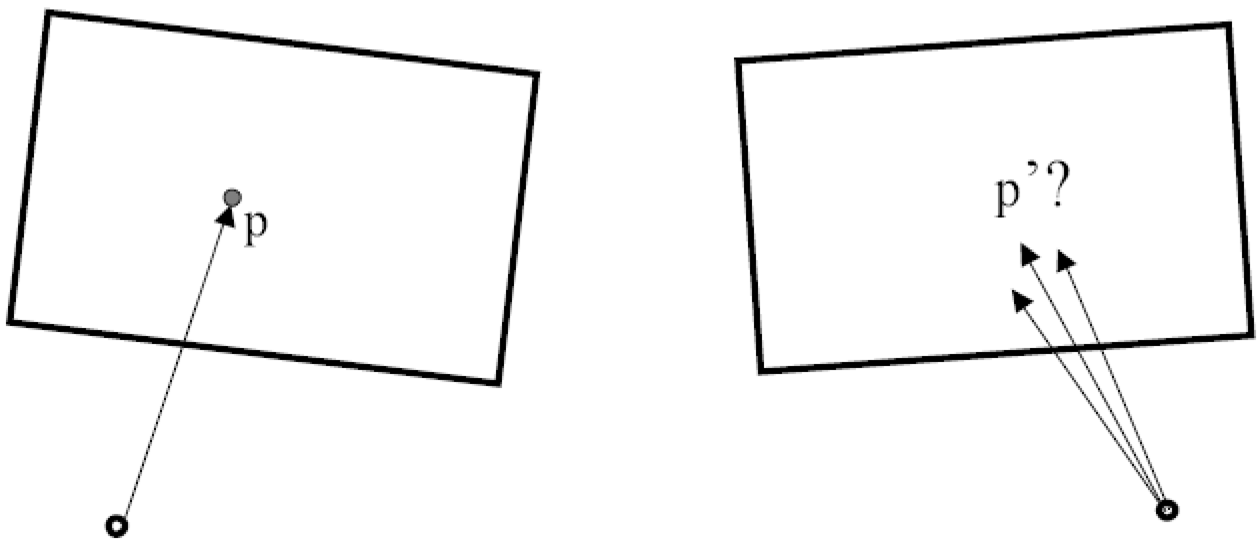
Stereo Correspondence Constraints



- Given p in the left image, where can the corresponding point p' in the right image be?

Slide credit: Kristen Grauman

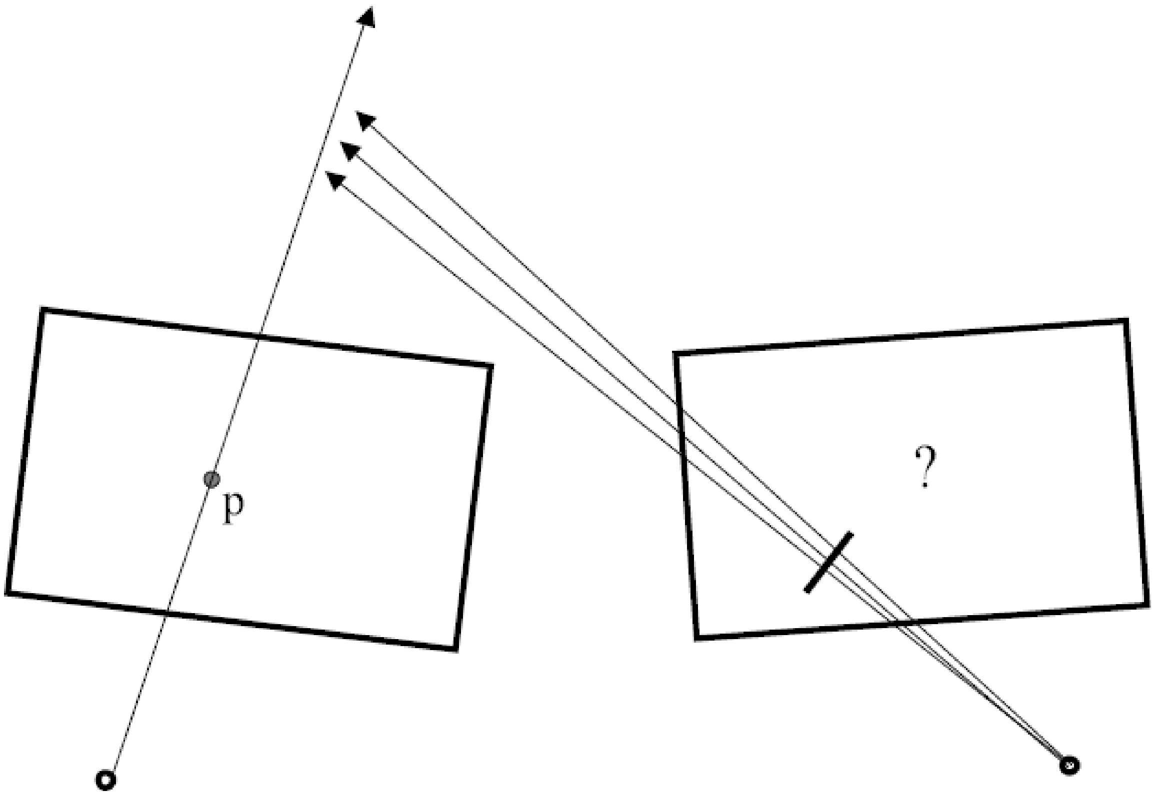
Stereo Correspondence Constraints



- Given p in the left image, where can the corresponding point p' in the right image be?

Slide credit: Kristen Grauman

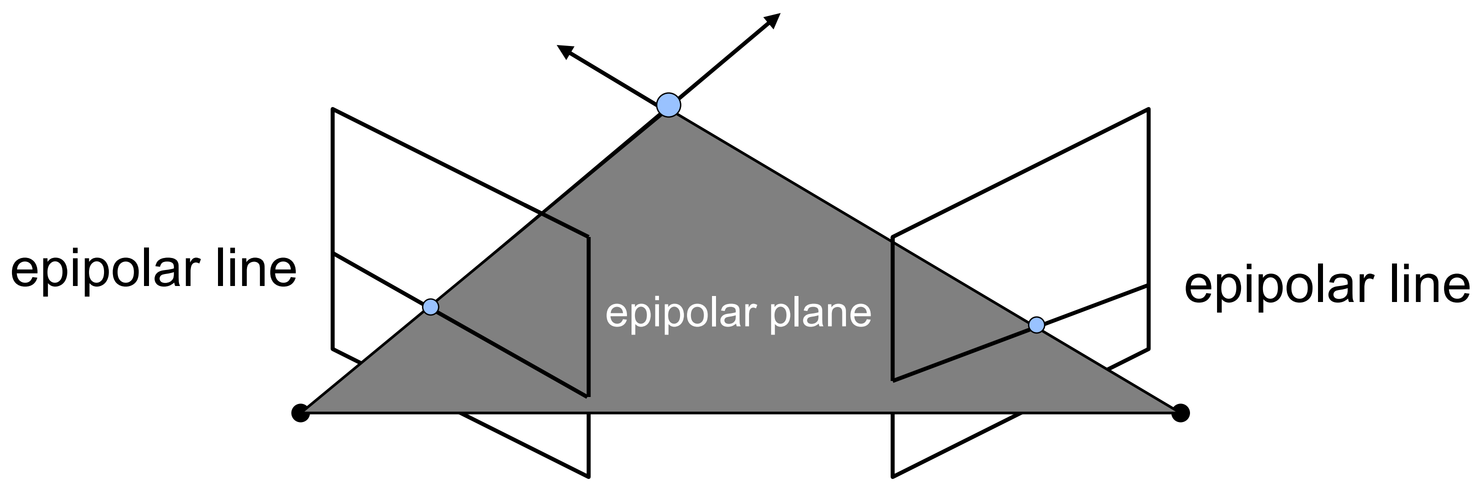
Stereo Correspondence Constraints



Slide credit: Kristen Grauman

Stereo Correspondence Constraints

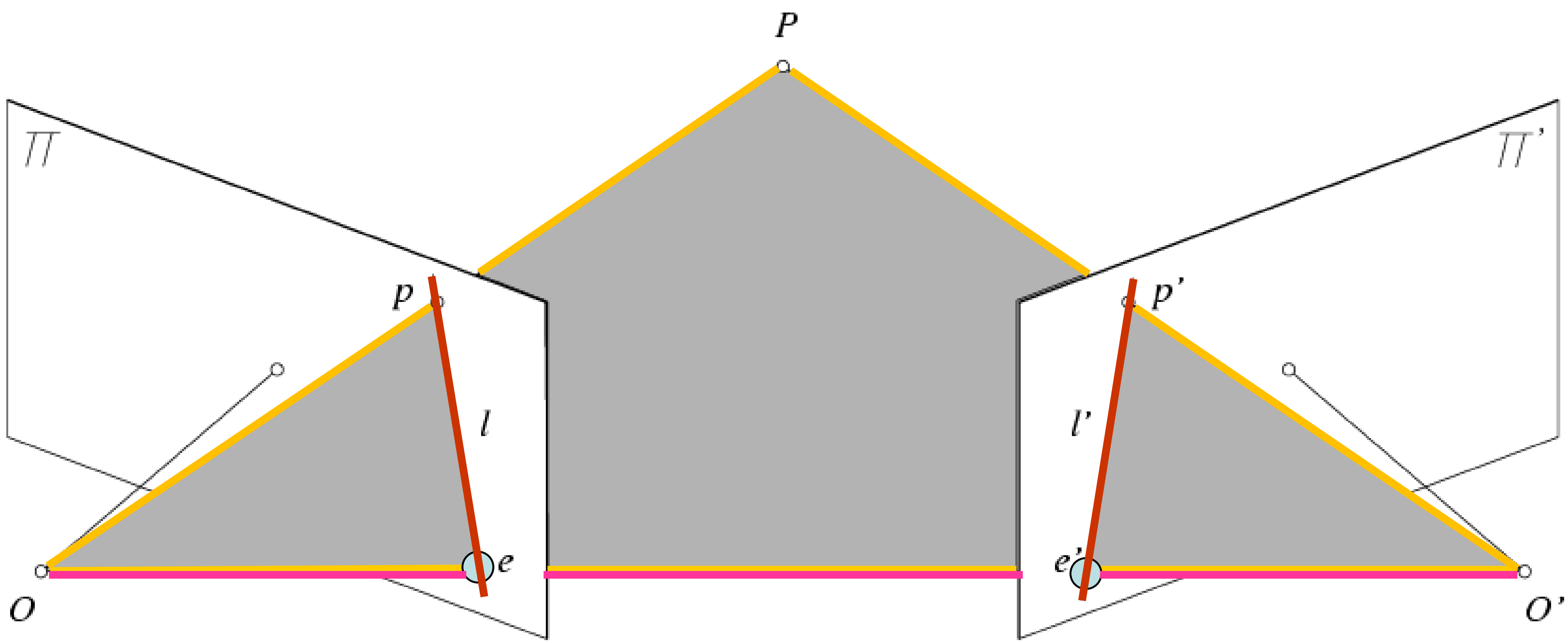
- Geometry of two views allows us to constrain where the corresponding pixel for some image point in the first view must occur in the second view.



- Epipolar constraint: Why is this useful?
  - Reduces correspondence problem to 1D search along conjugate epipolar lines.

Slide adapted from Steve Seitz

Epipolar Geometry



- Epipolar Plane
- Baseline
- Epipoles
- Epipolar Lines

Slide adapted from Marc Pollefeys

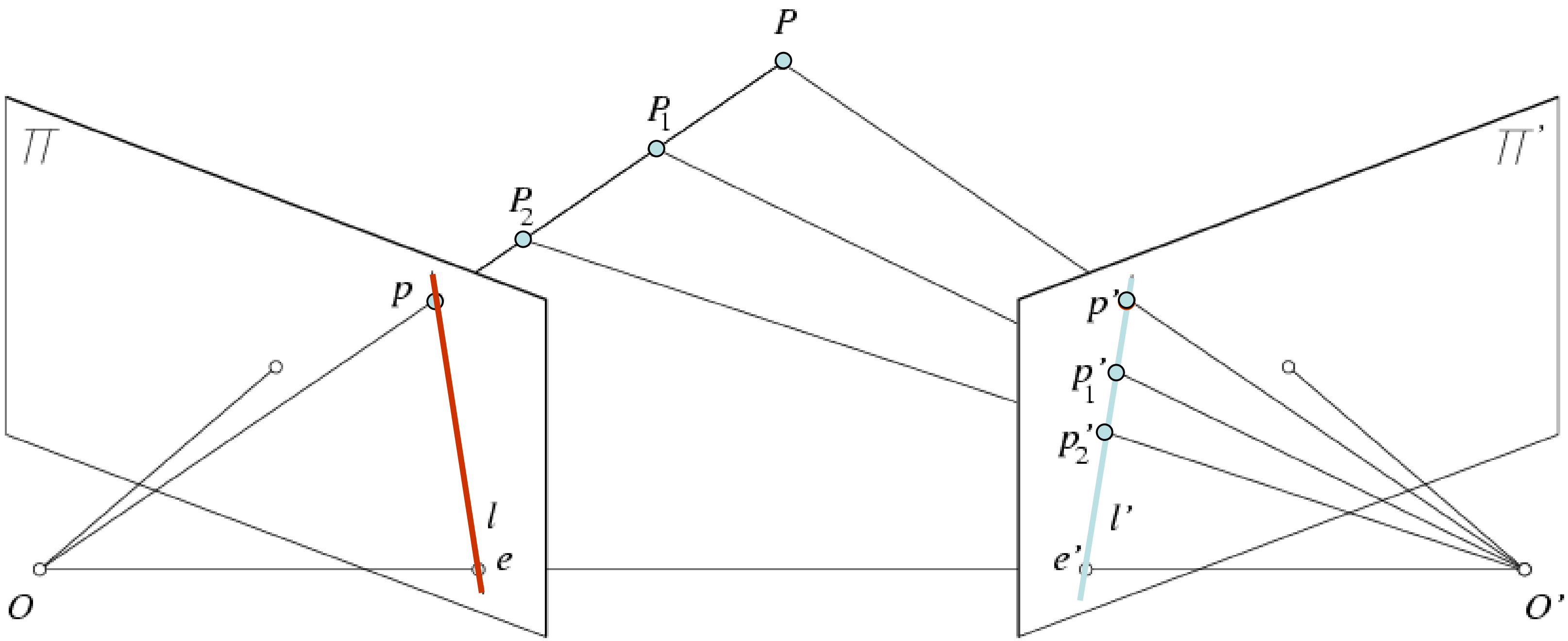
Epipolar Geometry: Terms

- *Baseline*
  - Line joining the camera centers
- *Epipole*
  - Point of intersection of baseline with the image plane
- *Epipolar plane*
  - Plane containing baseline and world point
- *Epipolar line*
  - Intersection of epipolar plane with the image plane

- Properties
  - All epipolar lines intersect at the epipole.
  - An epipolar plane intersects the left and right image planes in epipolar lines.

Slide credit: Marc Pollefeys

Epipolar Constraint

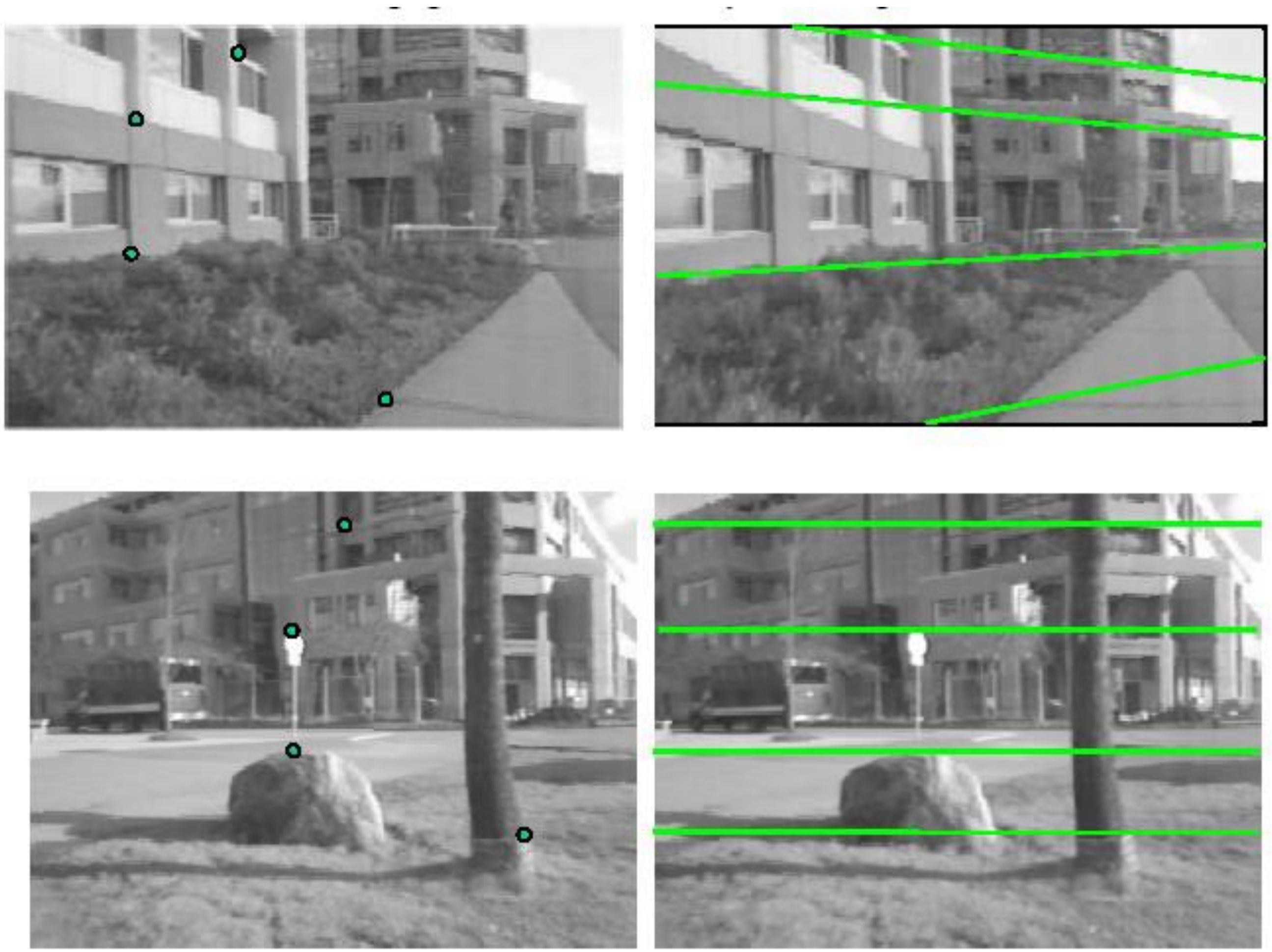


- Potential matches for p have to lie on the corresponding epipolar line l'.
- Potential matches for p' have to lie on the corresponding epipolar line l.

http://www.ai.sri.com/~luong/research/Meta3DViewer/EpipolarGeo.html

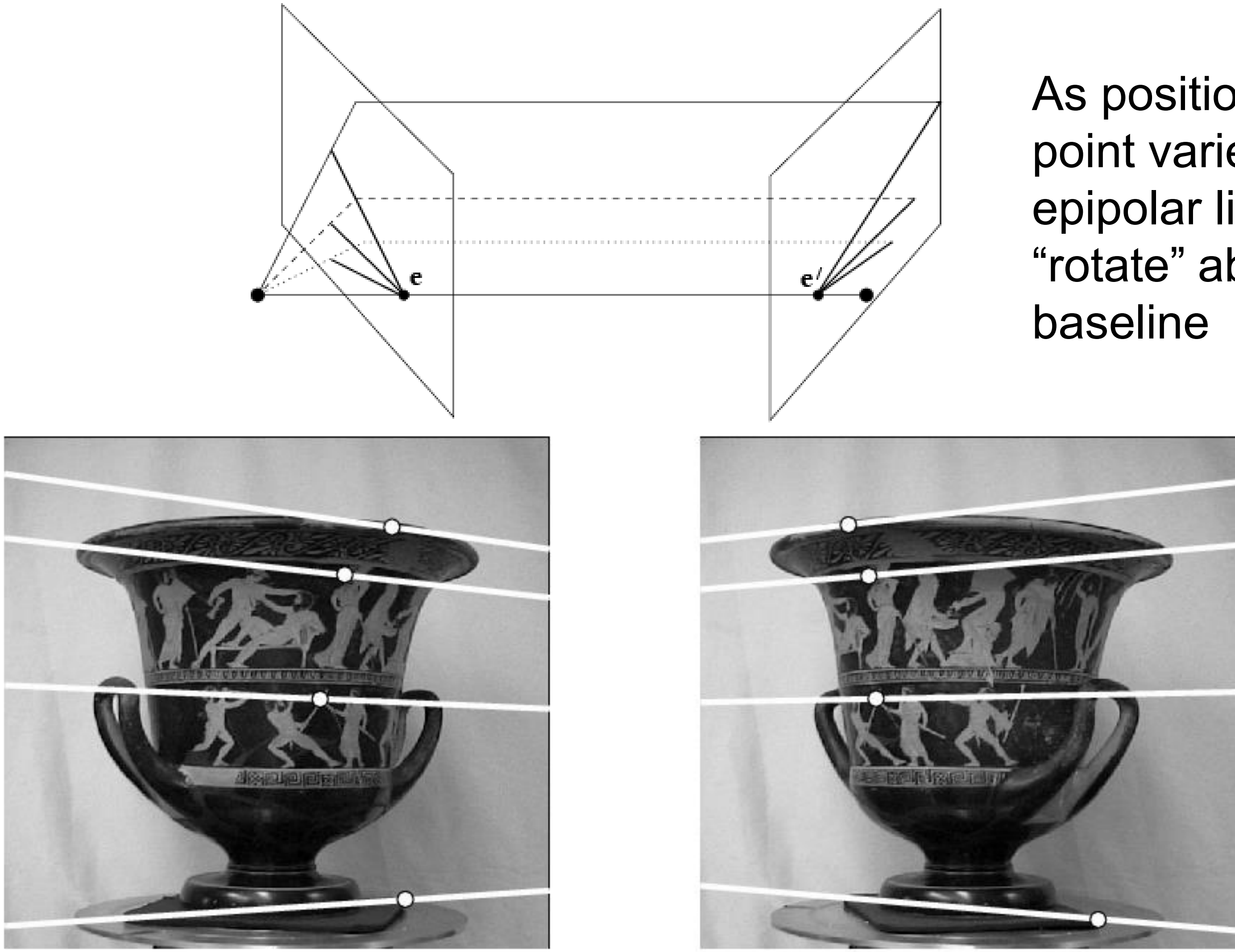
Slide credit: Marc Pollefeys

Example



Slide credit: Kristen Grauman

Example: Converging Cameras

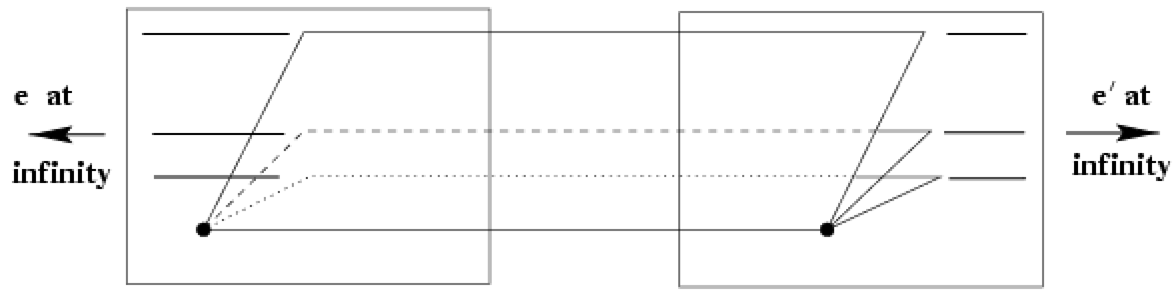


As position of 3D point varies, epipolar lines “rotate” about the baseline

Slide credit: Kristen Grauman

Figure from Hartley & Zisserman

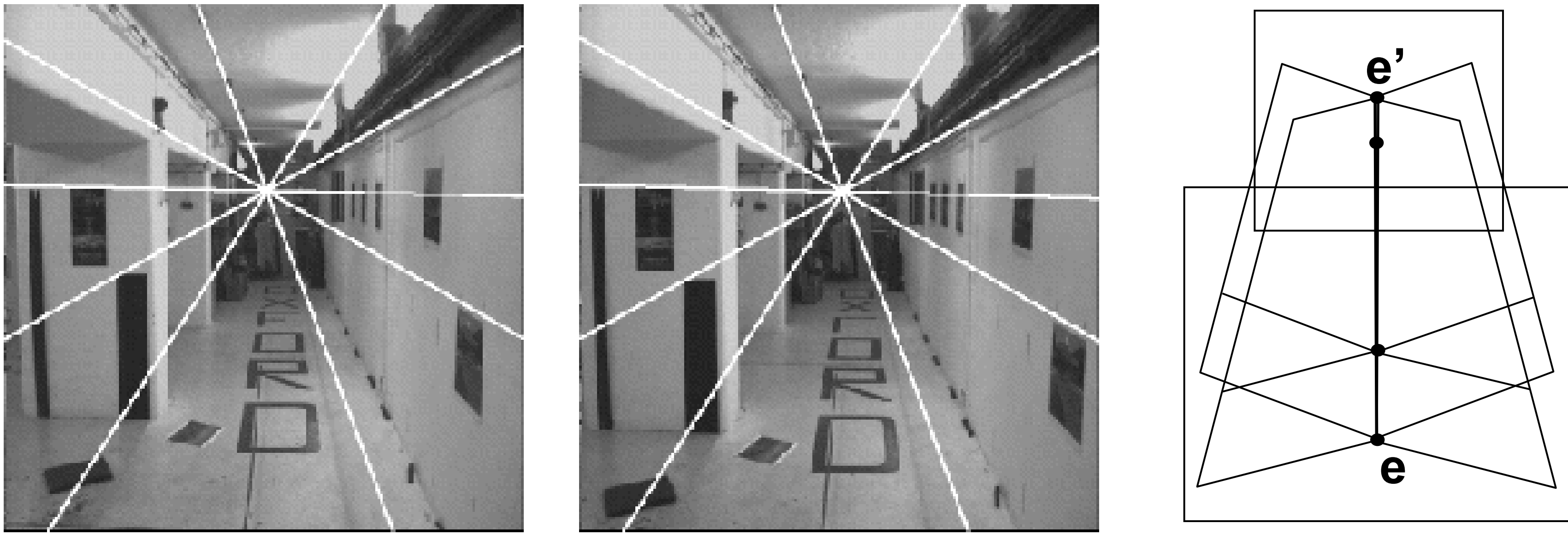
Example: Motion Parallel With Image Plane



Slide credit: Kristen Grauman

Figure from Hartley & Zisserman

Example: Forward Motion



- Epipole has same coordinates in both images.
- Points move along lines radiating from e: "Focus of expansion"

Slide credit: Kristen Grauman

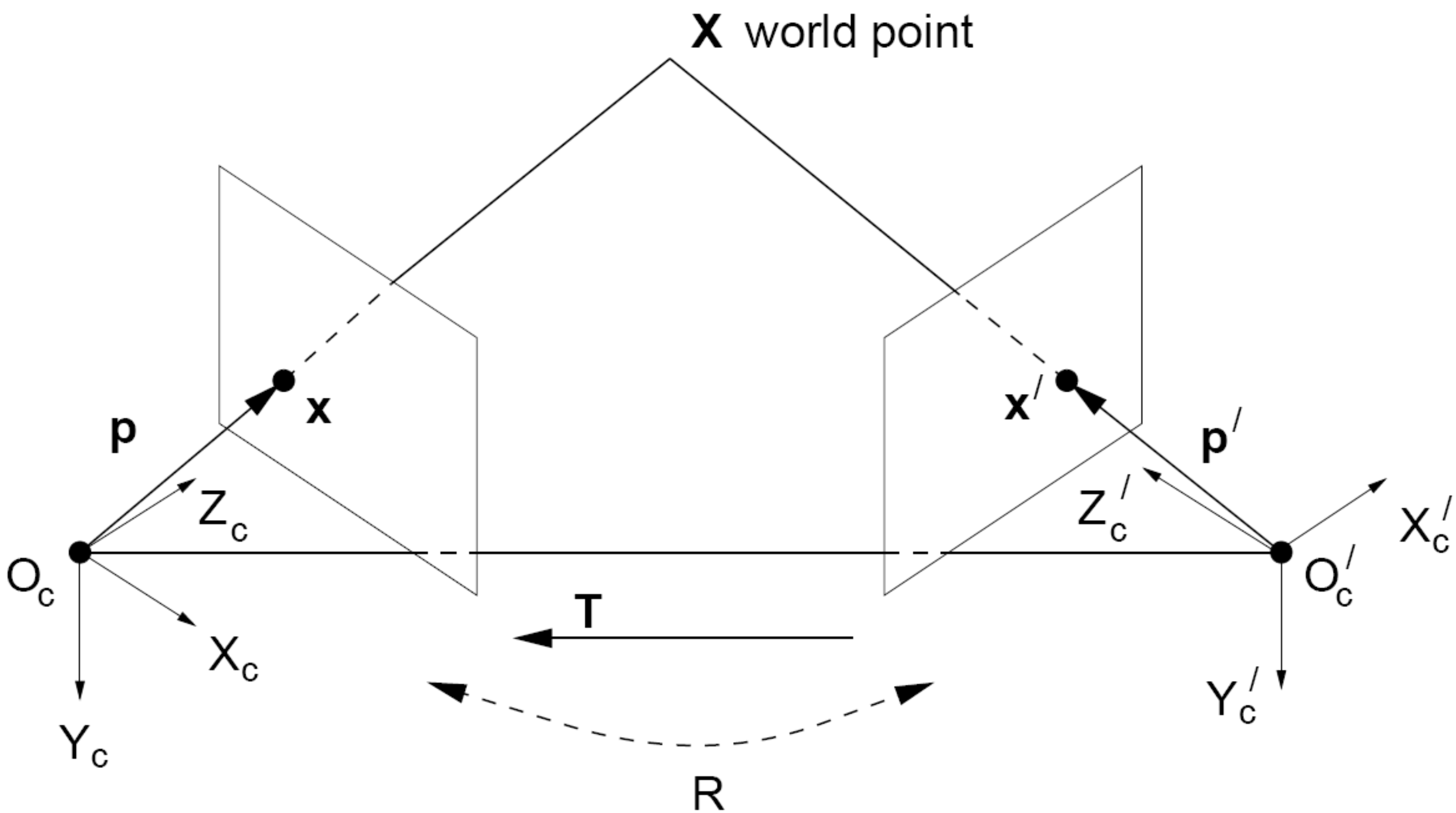
Figure from Hartley & Zisserman

Let’s Formalize This!

- For a given stereo rig, how do we express the epipolar constraints algebraically?

- For this, we will need some linear algebra.
  - But don’t worry! We’ll go through it step by step…

Stereo Geometry With Calibrated Cameras



- If the rig is calibrated, we know:
  - How to rotate and translate camera reference frame 1 to get to camera reference frame 2.
    - Rotation: 3 x 3 matrix; translation: 3 vector.

Slide credit: Kristen Grauman, Steve Seitz

Rotation Matrix

$$\mathbf{R}_x(\alpha) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\alpha & -\sin\alpha \\ 0 & \sin\alpha & \cos\alpha \end{bmatrix}$$

$$\mathbf{R}_y(\beta) = \begin{bmatrix} \cos\beta & 0 & \sin\beta \\ 0 & 1 & 0 \\ -\sin\beta & 0 & \cos\beta \end{bmatrix}$$

$$\mathbf{R}_z(\gamma) = \begin{bmatrix} \cos\gamma & -\sin\gamma & 0 \\ \sin\gamma & \cos\gamma & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Express 3D rotation as series of rotations around coordinate axes by angles α, β, γ

Overall rotation is product of these elementary rotations:

$$\mathbf{R} = \mathbf{R}_x\mathbf{R}_y\mathbf{R}_z$$

Computer Vision

Slide credit: Kristen Grauman

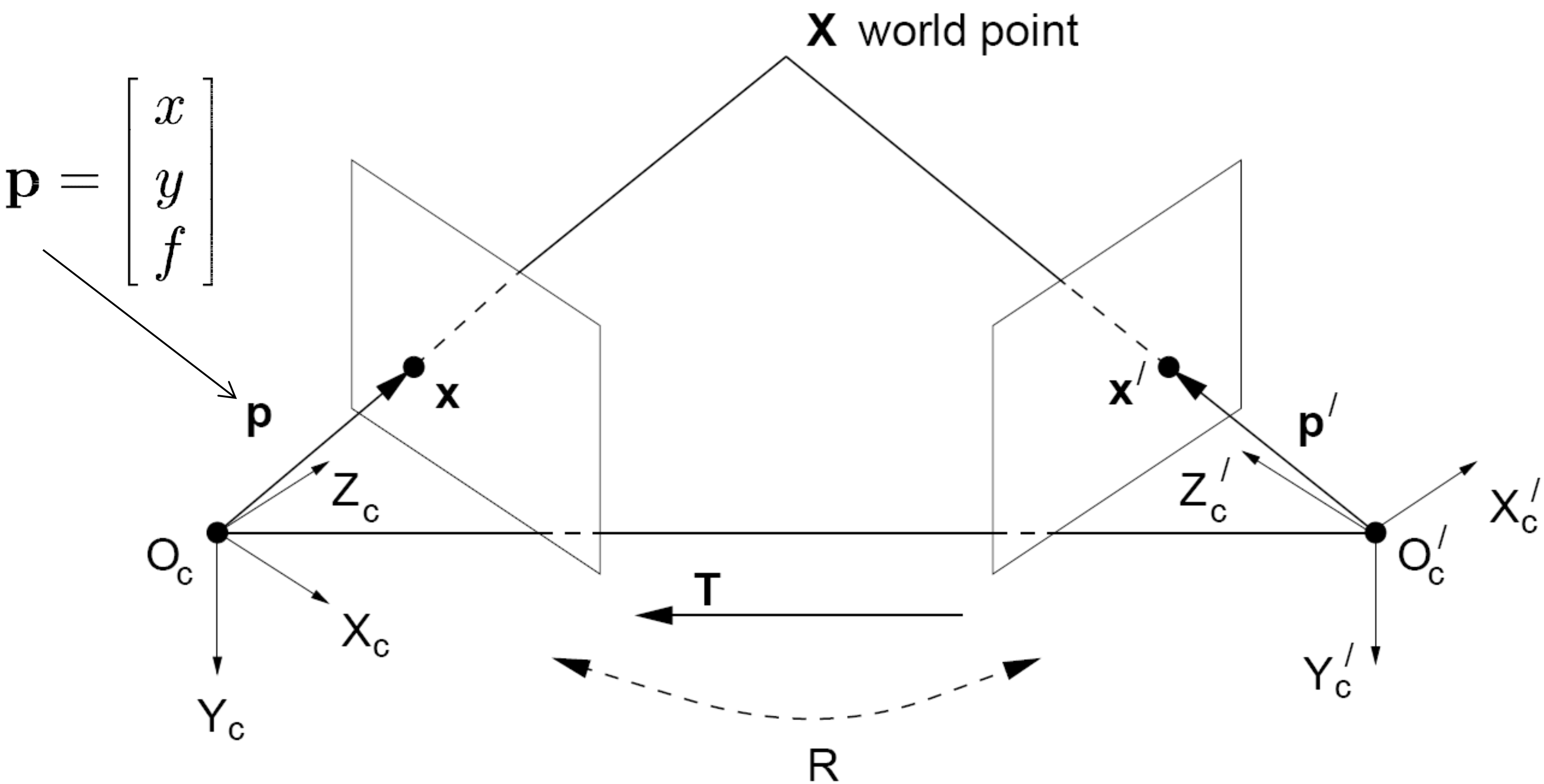
3D Rigid Transformation

$$\begin{bmatrix} X' \\ Y' \\ Z' \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} + \begin{bmatrix} T_x \\ T_y \\ T_z \end{bmatrix}$$

$$\mathbf{X}' = \mathbf{R}\mathbf{X} + \mathbf{T}$$

Slide credit: Kristen Grauman

Stereo Geometry With Calibrated Cameras



- Camera-centered coordinate systems are related by known rotation R and translation T:

$$\mathbf{X}' = \mathbf{R}\mathbf{X} + \mathbf{T}$$

Slide credit: Kristen Grauman

Excursion: Cross Product

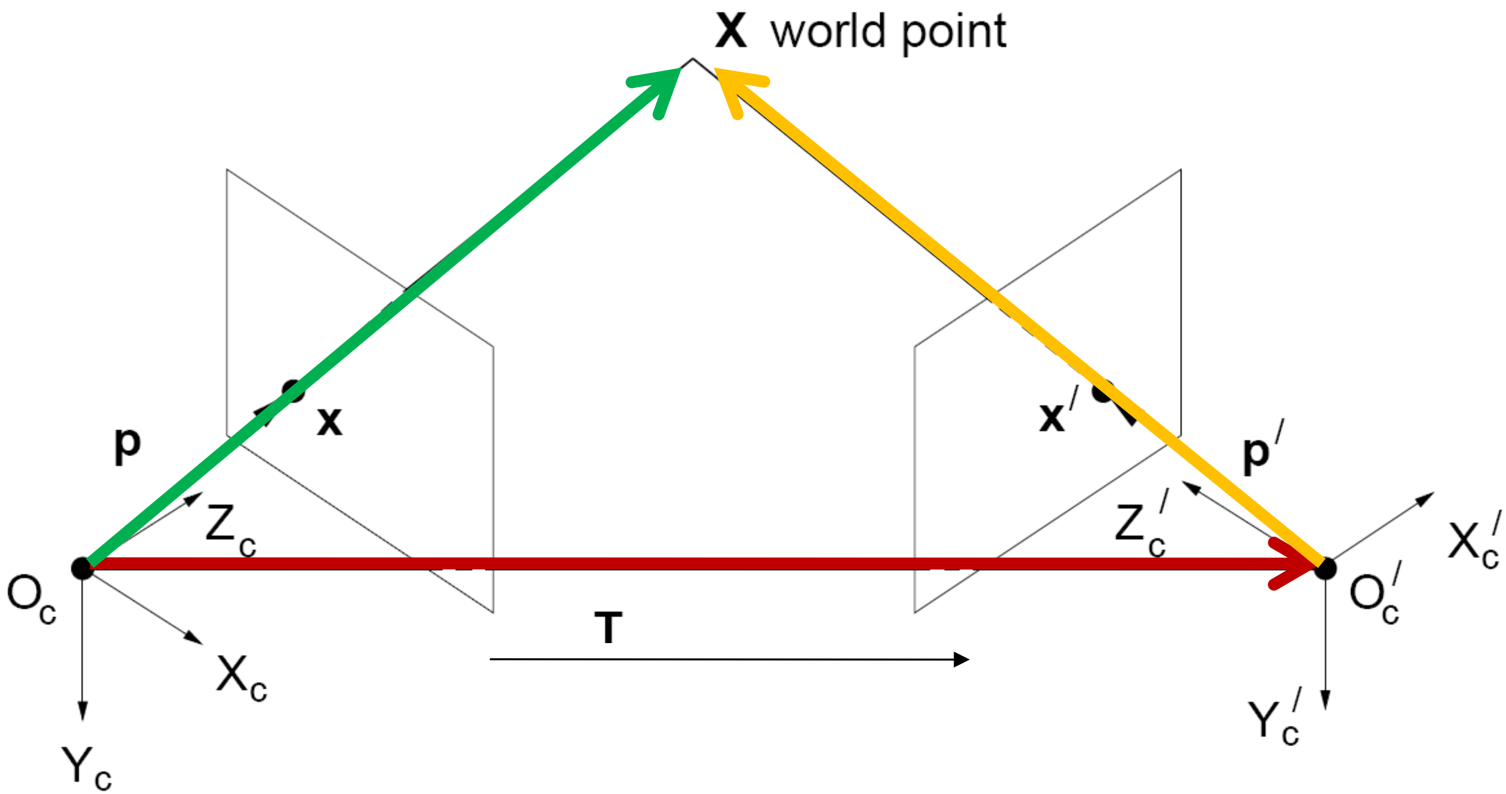
$$\vec{a} \times \vec{b} = \vec{c}$$

$$\begin{aligned} \vec{a} \cdot \vec{c} &= 0 \\ \vec{b} \cdot \vec{c} &= 0 \end{aligned}$$

- Vector cross product takes two vectors and returns a third vector that's perpendicular to both inputs.
- So here, c is perpendicular to both a and b, which means the dot product is 0.

Slide credit: Kristen Grauman

From Geometry to Algebra



$$\boxed{\mathbf{X}'} = \mathbf{R}\boxed{\mathbf{X}} + \boxed{\mathbf{T}}$$

$$\underbrace{\mathbf{T} \times \mathbf{X}'}_{\text{Normal to the plane}} = \mathbf{T} \times \mathbf{R}\mathbf{X} + \mathbf{T} \times \mathbf{T}$$

$$= \mathbf{T} \times \mathbf{R}\mathbf{X}$$

$$\mathbf{X}' \cdot \left(\mathbf{T} \times \mathbf{X}'\right) = \mathbf{X}' \cdot \left(\mathbf{T} \times \mathbf{R}\mathbf{X}\right)$$

$$0 = \mathbf{X}' \cdot \left(\mathbf{T} \times \mathbf{R}\mathbf{X}\right)$$

Slide credit: Kristen Grauman

Matrix Form of Cross Product

$$\vec{a} \times \vec{b} \qquad\qquad = \vec{c} \qquad \begin{aligned} \vec{a} \cdot \vec{c} &= 0 \\ \vec{b} \cdot \vec{c} &= 0 \end{aligned}$$

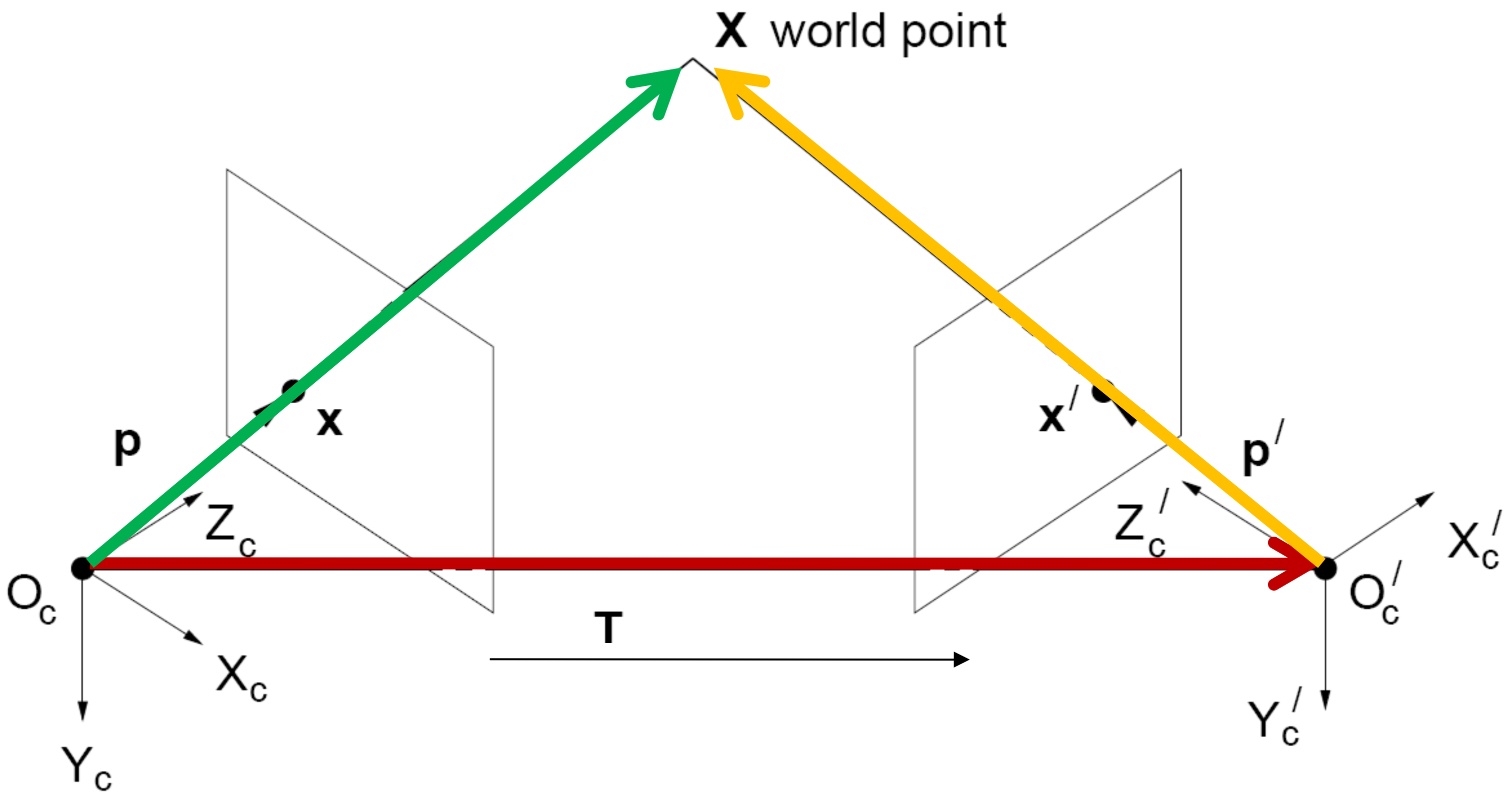
"skew symmetric" matrix

$$[a_\times] = \begin{bmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{bmatrix}$$

$$\boxed{\vec{a} \times \vec{b} = [a_\times]\vec{b}}$$

Slide credit: Kristen Grauman

From Geometry to Algebra



$$\boxed{\mathbf{X'}} = \mathbf{R}\boxed{\mathbf{X}} + \boxed{\mathbf{T}}$$

$$\underbrace{\mathbf{T} \times \mathbf{X'}}_{\text{Normal to the plane}} = \mathbf{T} \times \mathbf{RX} + \mathbf{T} \times \mathbf{T}$$

$$= \mathbf{T} \times \mathbf{RX}$$

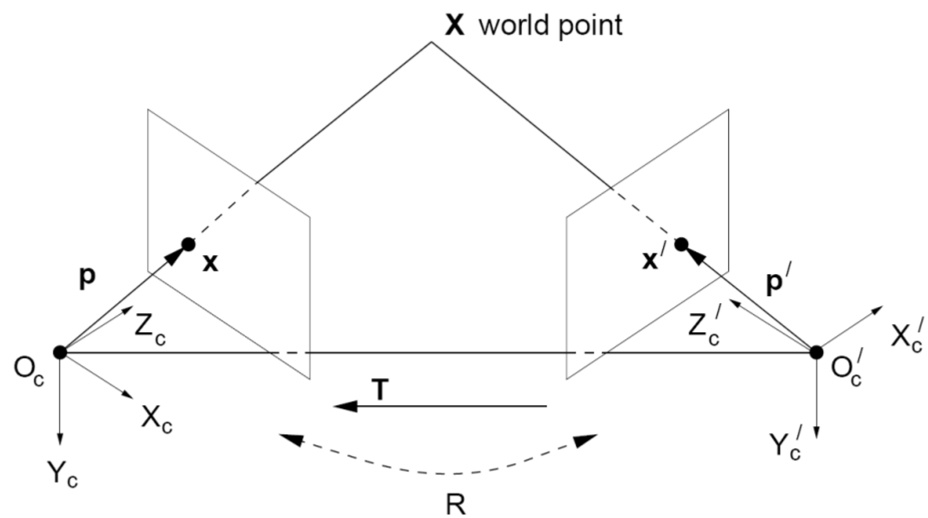
$$\mathbf{X'} \cdot (\mathbf{T} \times \mathbf{X'}) = \mathbf{X'} \cdot (\mathbf{T} \times \mathbf{RX})$$

$$\boxed{0 = \mathbf{X'} \cdot (\mathbf{T} \times \mathbf{RX})}$$

Essential Matrix

$$\boxed{\mathbf{X}' \cdot (\mathbf{T} \times \mathbf{R}\mathbf{X}) = 0}$$

$$\mathbf{X}' \cdot (\mathbf{T}_x \ \mathbf{R}\mathbf{X}) = 0$$



Let $\mathbf{E} = \mathbf{T}_x \mathbf{R}$

$$\mathbf{X}'^T \mathbf{E} \mathbf{X} = 0$$

- This holds for the rays p and p' that are parallel to the camera-centered position vectors X and X', so we have:

$$\boxed{\mathbf{p}'^T \mathbf{E} \mathbf{p} = 0}$$

- $\mathbf{E}$ is called the Essential matrix, which relates corresponding image points [Longuet-Higgins 1981]

Slide credit: Kristen Grauman

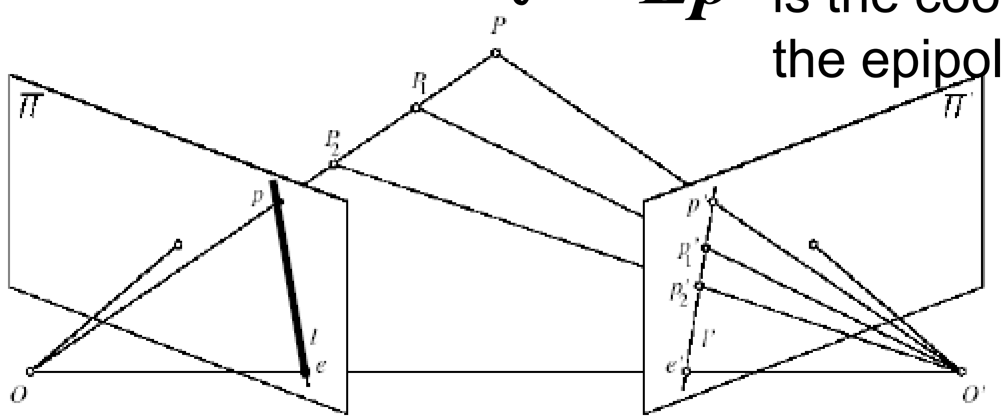
Essential Matrix and Epipolar Lines

$$\boxed{\mathbf{p'}^{\mathrm{T}}\mathbf{E}\mathbf{p} = 0}$$

Epipolar constraint: if we observe point p in one image, then its position p' in second image must satisfy this equation.

$\boldsymbol{l}' = \boldsymbol{E}\boldsymbol{p}$ is the coordinate vector representing the epipolar line for point p

(i.e., the line is given by: $l'^{\top}\mathrm{x} = 0$)



$\boldsymbol{l} = \boldsymbol{E}^T\boldsymbol{p}'$ is the coordinate vector representing the epipolar line for point p'

Slide credit: Kristen Grauman

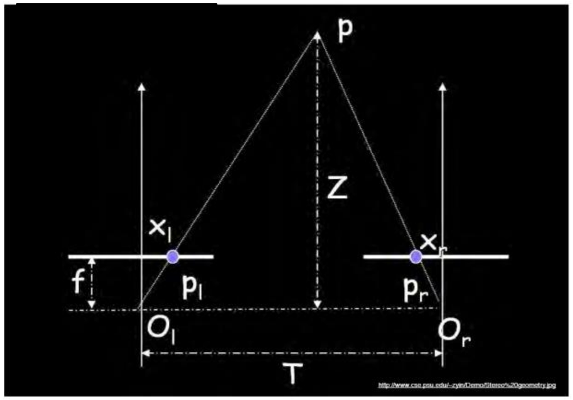
Essential Matrix: Properties

- Relates image of corresponding points in both cameras, given rotation and translation.
- Assuming intrinsic parameters are known

$$\mathbf{E} = \mathbf{T}_x \mathbf{R}$$

Slide credit: Kristen Grauman

Essential Matrix Example: Parallel Cameras



$$\mathbf{R} = \mathbf{I}$$

$$\mathbf{T} = [-d, 0, 0]^{\mathrm{T}}$$

$$\mathbf{E} = [\mathbf{T_x}]\mathbf{R} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & d \\ 0 & -d & 0 \end{bmatrix}$$

$$\mathbf{p}'^{\mathrm{T}}\mathbf{E}\mathbf{p} = 0$$

$$\begin{bmatrix} x' & y' & f \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & d \\ 0 & -d & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ f \end{bmatrix} = 0$$

$$\Leftrightarrow \begin{bmatrix} x' & y' & f \end{bmatrix} \begin{bmatrix} 0 \\ df \\ -dy \end{bmatrix} = 0$$

$$\boxed{\Leftrightarrow y = y'}$$

For the parallel cameras, image of any point must lie on same horizontal line in each image plane.

Slide credit: Kristen Grauman

More General Case

Image $I(x, y)$ | Disparity map $D(x, y)$ | Image $I'(x', y')$

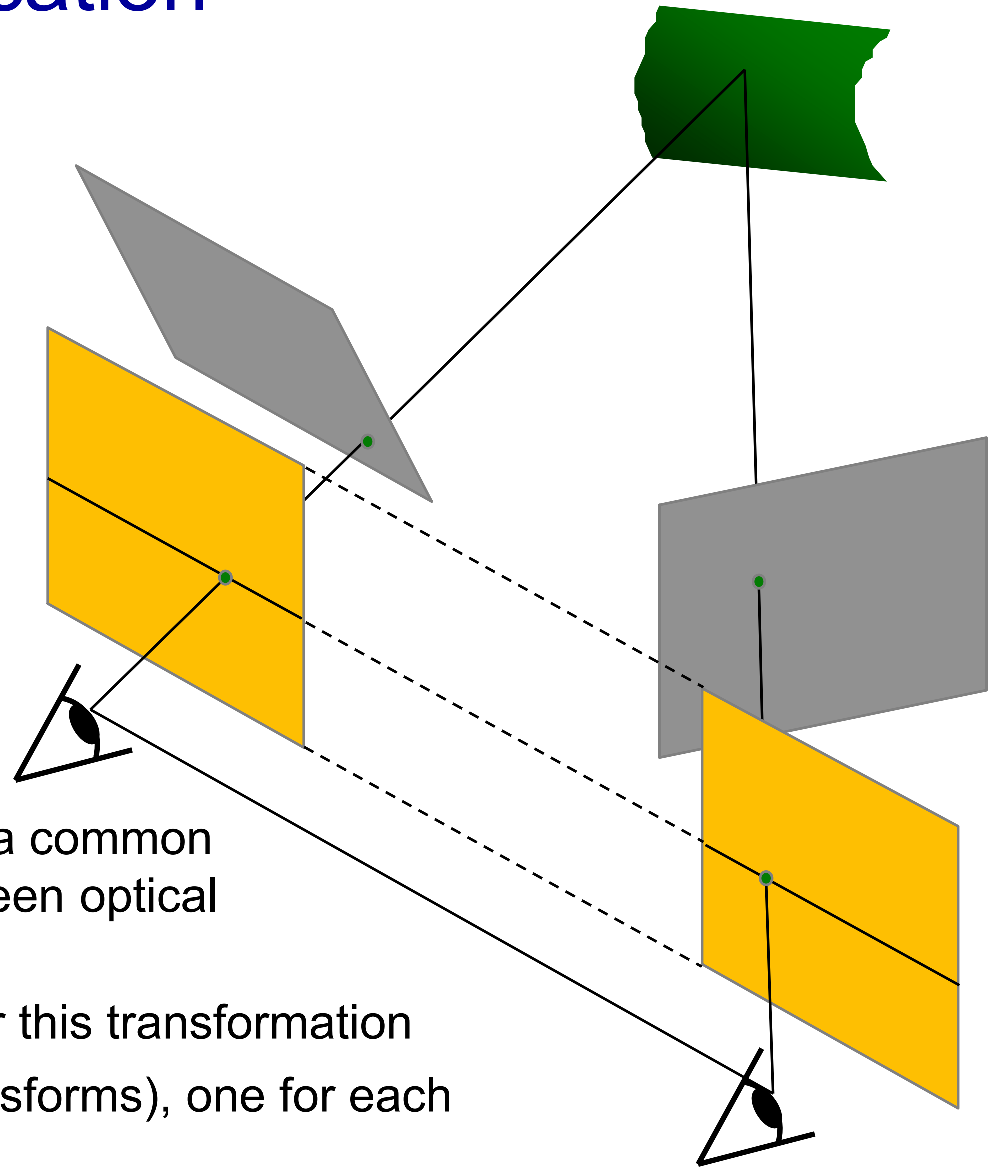
$$(x', y') = (x + D(x, y), y)$$

What about when cameras' optical axes are not parallel?

Slide credit: Kristen Grauman

Stereo Image Rectification

- In practice, it is convenient if image scanlines are the epipolar lines.

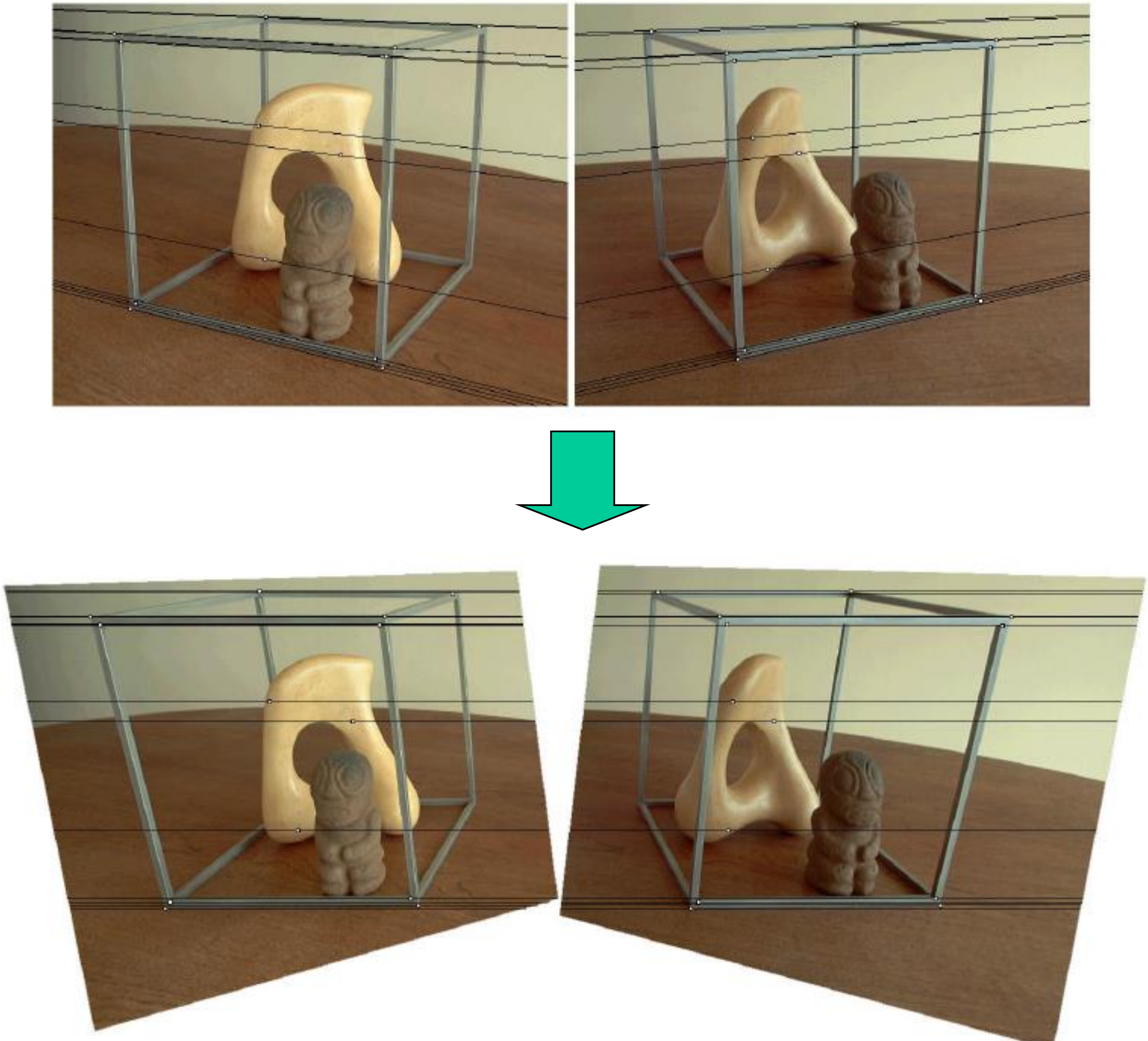


- Algorithm
  - Reproject image planes onto a common plane parallel to the line between optical centers
  - Pixel motion is horizontal after this transformation
  - Two homographies (3×3 transforms), one for each input image reprojection

Slide adapted from Li Zhang

C. Loop & Z. Zhang, Computing Rectifying Homographies for Stereo Vision. CVPR'99

Stereo Image Rectification: Example



Source: Alyosha Efros

Topics of This Lecture

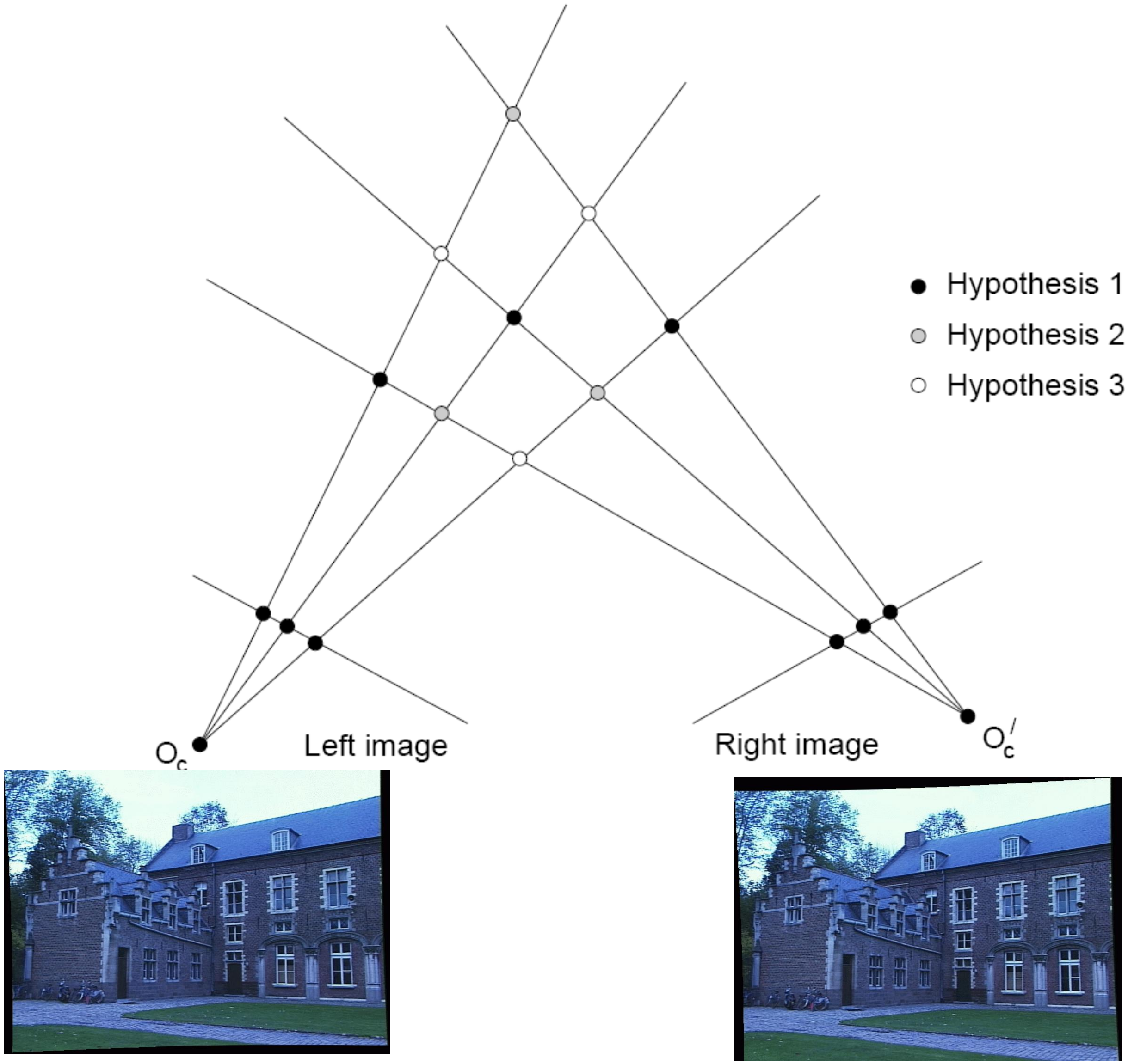
- Keypoint Localization
  - Recap: Interest points
  - Harris detector
- Local Descriptors
  - SIFT descriptor
- Epipolar geometry
  - Depth with stereo
  - Geometry for a simple stereo system
  - Case example with parallel optical axes
  - General case with calibrated cameras
- Stereopsis & 3D Reconstruction
  - Correspondence search
  - Additional correspondence constraints

Stereo Reconstruction

- Main Steps
  - Calibrate cameras
  - Rectify images
  - Compute disparity
  - Estimate depth

Slide credit: Kristen Grauman

Correspondence Problem

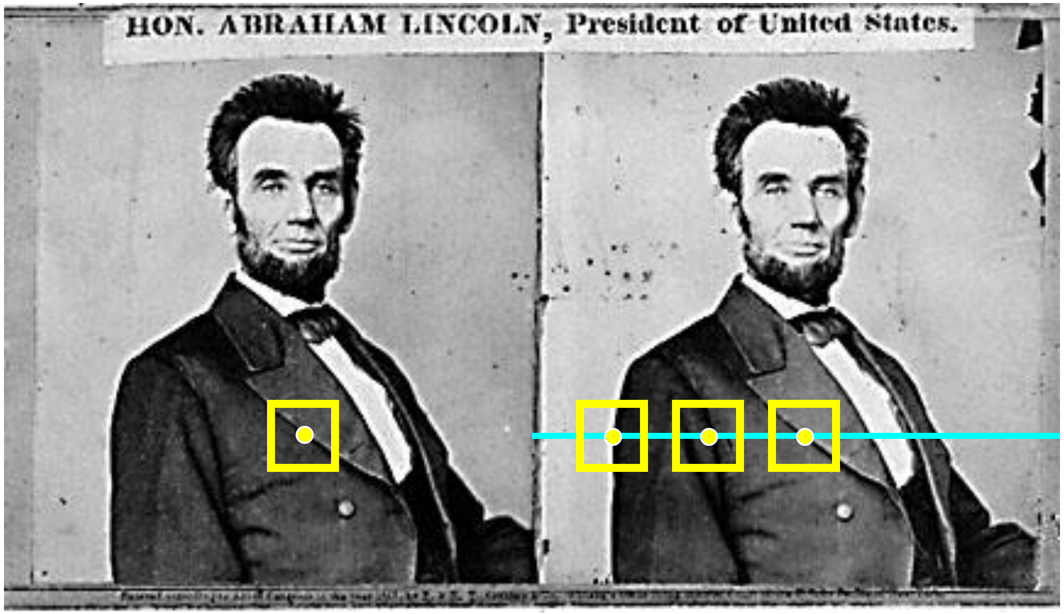


Multiple match hypotheses satisfy epipolar constraint, but which is correct?

Slide credit: Kristen Grauman

Figure from Gee & Cipolla 1999

Dense Correspondence Search



- For each pixel in the first image
  - Find corresponding epipolar line in the right image
  - Examine all pixels on the epipolar line and pick the best match (e.g. SSD, correlation)
  - Triangulate the matches to get depth information
- This is easiest when epipolar lines are scanlines
  - $\Rightarrow$ Rectify images first

adapted from Svetlana Lazebnik, Li Zhang

Example: Window Search

- Data from University of Tsukuba

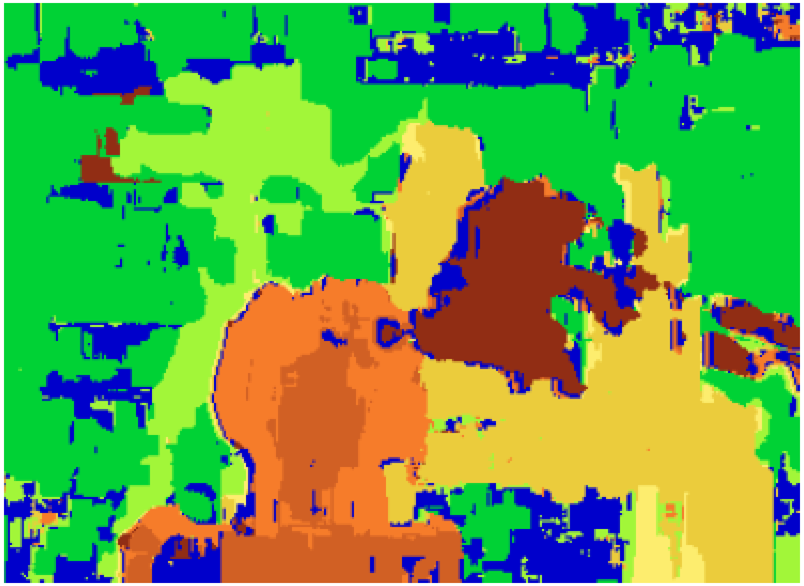
Scene

Ground truth

Slide credit: Kristen Grauman

Example: Window Search

- Data from University of Tsukuba



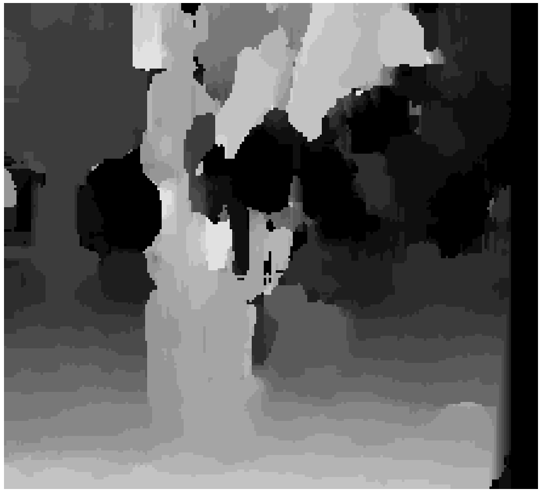
Window-based matching
(best window size)

Ground truth

Slide credit: Kristen Grauman

Effect of Window Size

$W = 3$



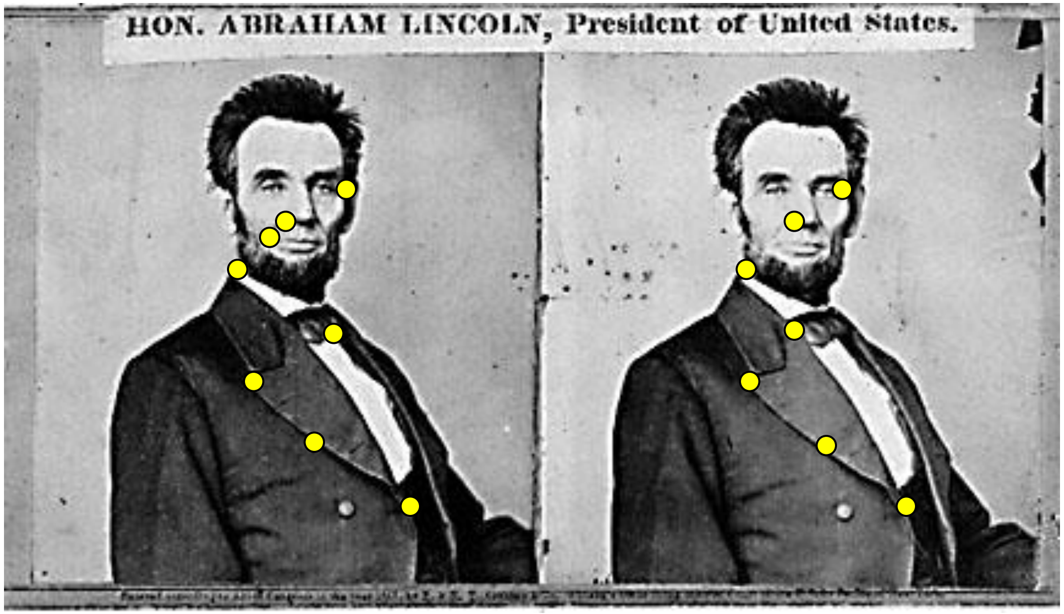
$W = 20$

Want window large enough to have sufficient intensity variation, yet small enough to contain only pixels with about the same disparity.

Slide credit: Kristen Grauman

Figures from Li Zhang

Alternative: Sparse Correspondence Search



- Idea: Restrict search to sparse set of detected features
- Rather than pixel values (or lists of pixel values) use *feature descriptor* and an associated *feature distance*
- Still narrow search further by epipolar geometry

What would make good features?

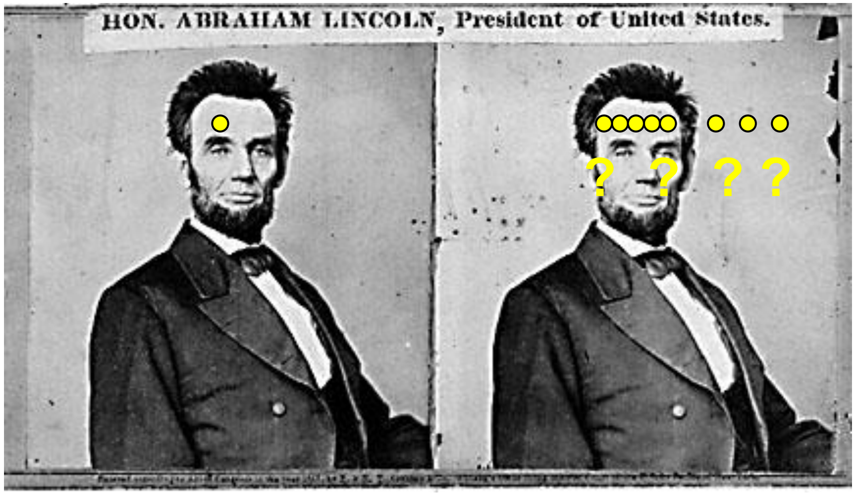
Slide credit: Kristen Grauman

Dense vs. Sparse

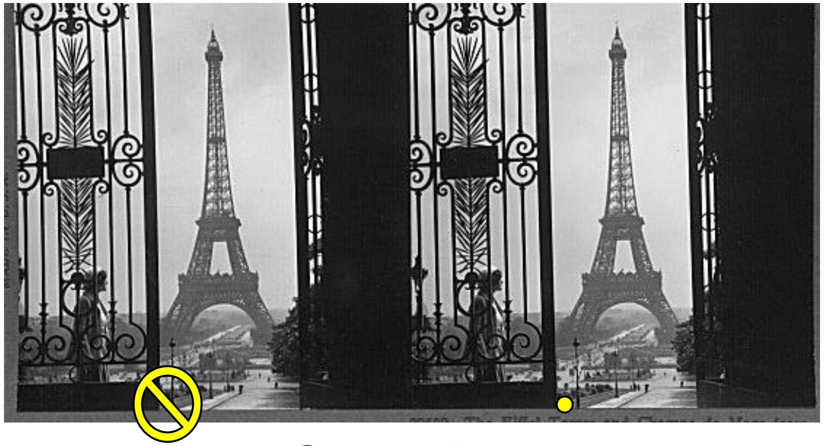
- Sparse
  - Efficiency
  - Can have more reliable feature matches, less sensitive to illumination than raw pixels
  - But…
    - Have to know enough to pick good features
    - Sparse information
- Dense
  - Simple process
  - More depth estimates, can be useful for surface reconstruction
  - But…
    - Breaks down in textureless regions anyway
    - Raw pixel distances can be brittle
    - Not good with very different viewpoints

Slide credit: Kristen Grauman

Difficulties in Similarity Constraint



Untextured surfaces



Occlusions

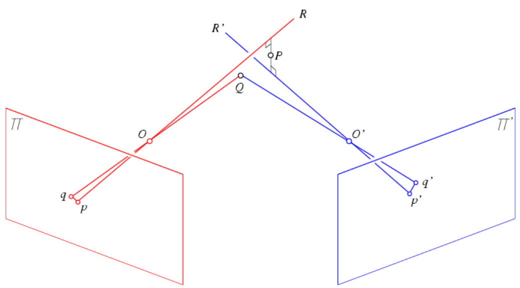
Slide credit: Kristen Grauman

Summary: Stereo Reconstruction

- Main Steps
  - Calibrate cameras
  - Rectify images
  - Compute disparity
  - Estimate depth

- So far, we have only considered calibrated cameras…

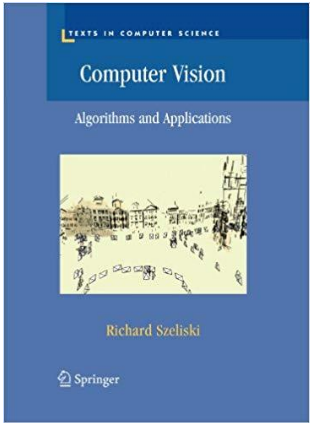
- Next lecture
  - Uncalibrated cameras
  - Camera parameters
  - Revisiting epipolar geometry
  - Robust fitting



Slide credit: Kristen Grauman

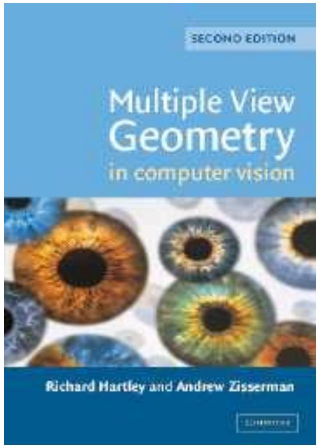
References and Further Reading

- Background information on epipolar geometry and stereopsis can be found in Chapter 11 of



R. Szeliski
Computer Vision – Algorithms and Applications
Springer, 2010
([draft version available online](#))

- More detailed information (if you really want to implement 3D reconstruction algorithms) can be found in Chapters 9 and 10 of



R. Hartley, A. Zisserman
Multiple View Geometry in Computer Vision
2nd Ed., Cambridge Univ. Press, 2004